

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168

Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : height

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : height

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

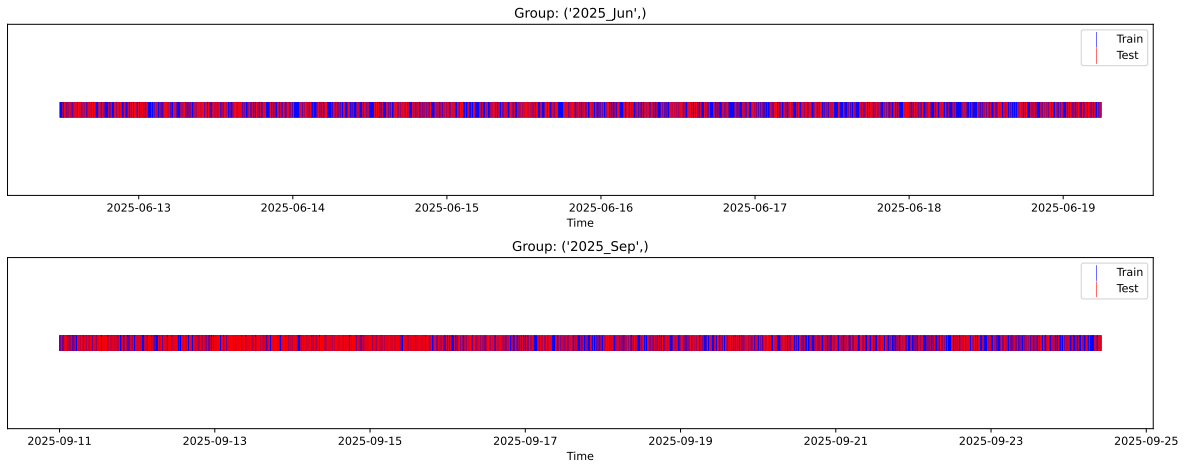
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

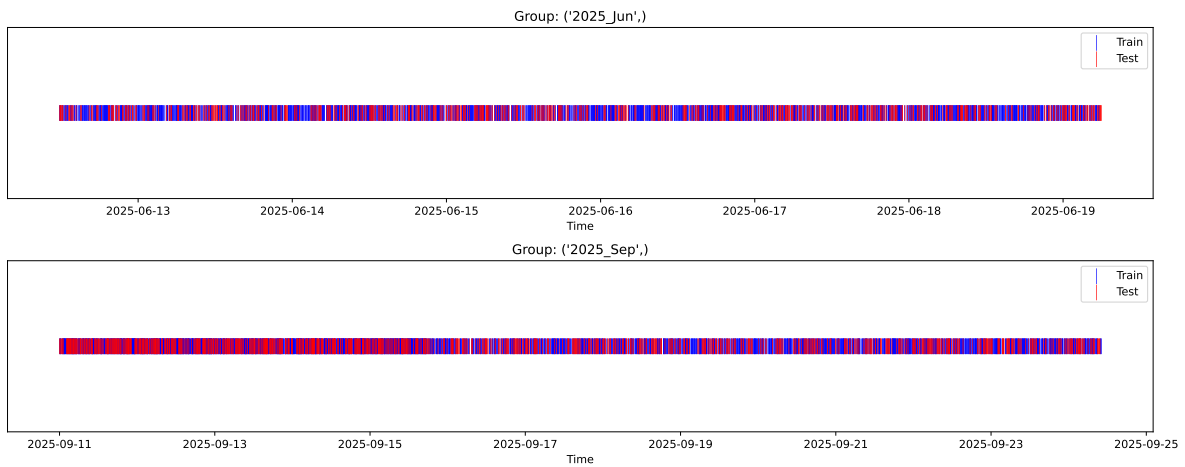
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

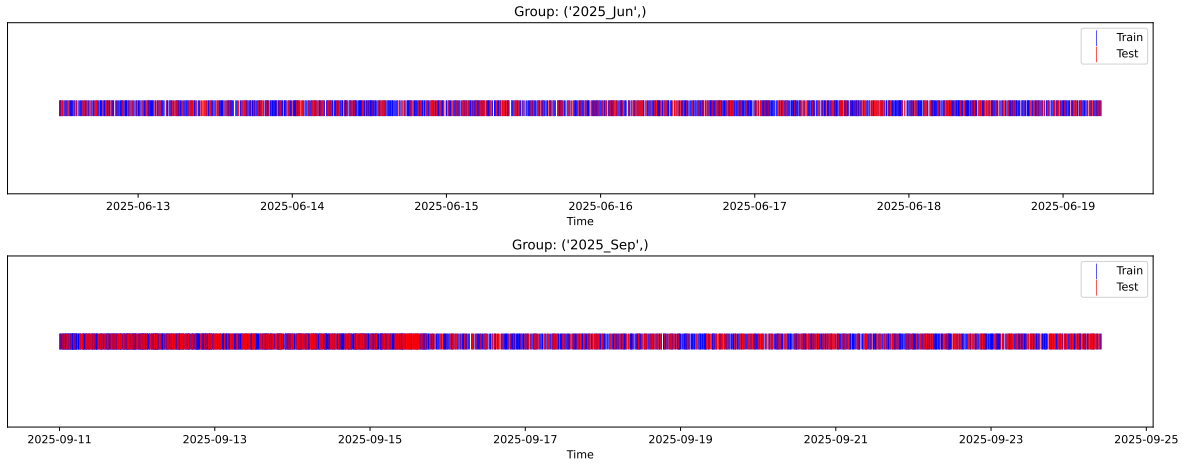
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



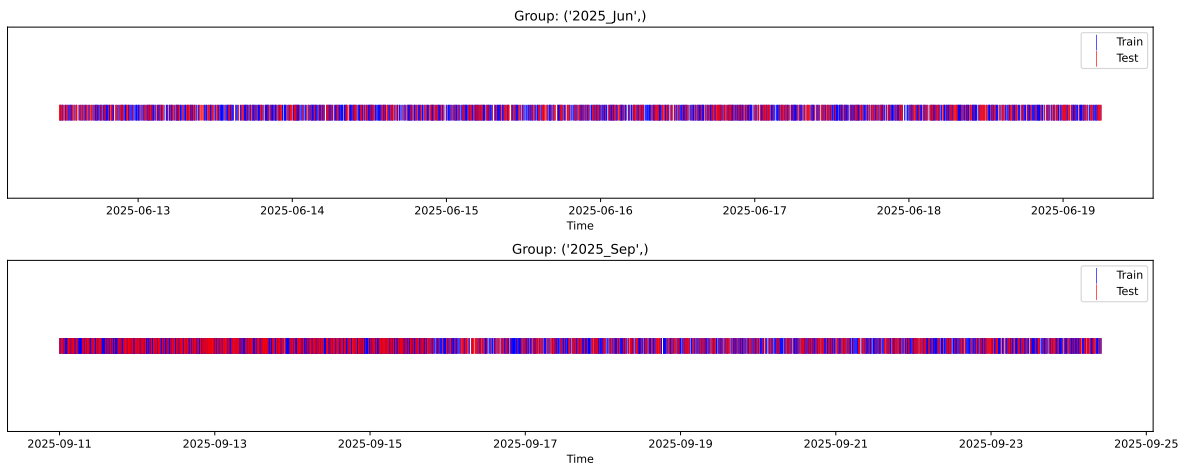
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



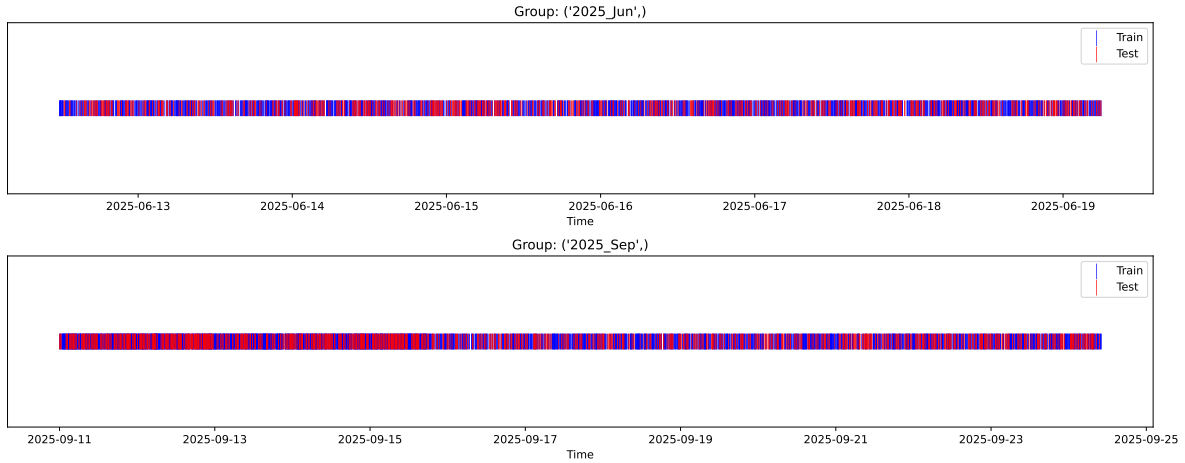
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



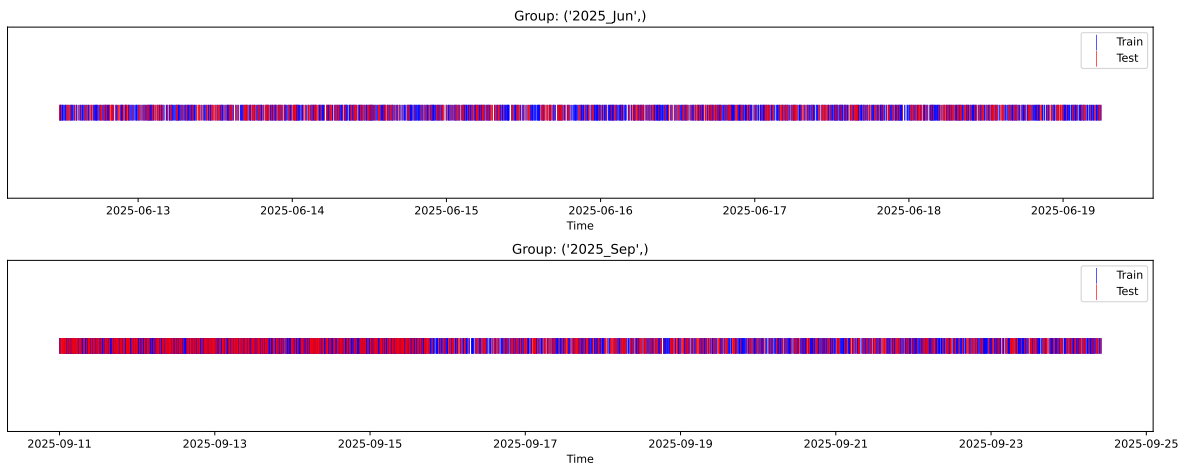
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



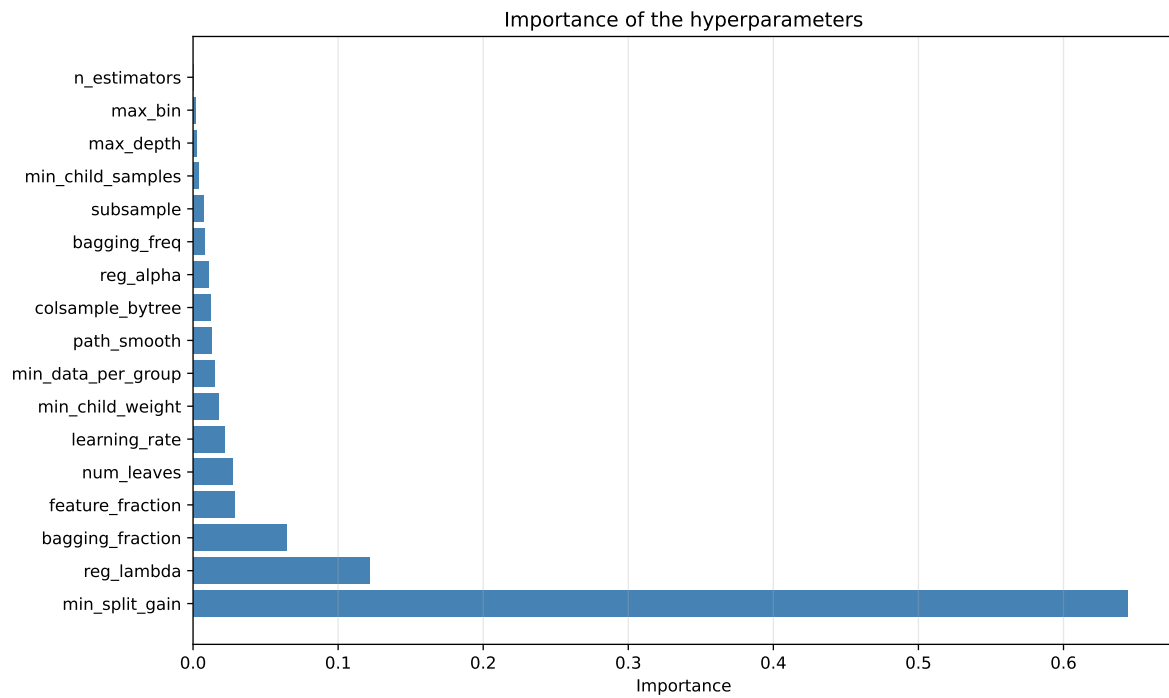
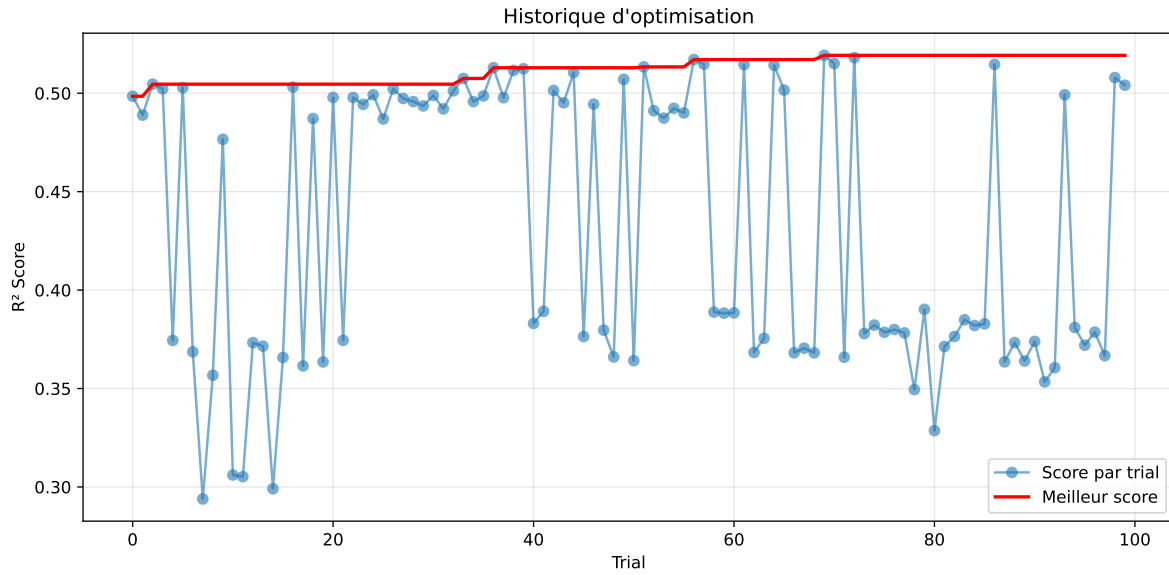
2.3 Model : LGBM

2.3.1 Gridsearch — Group: 155

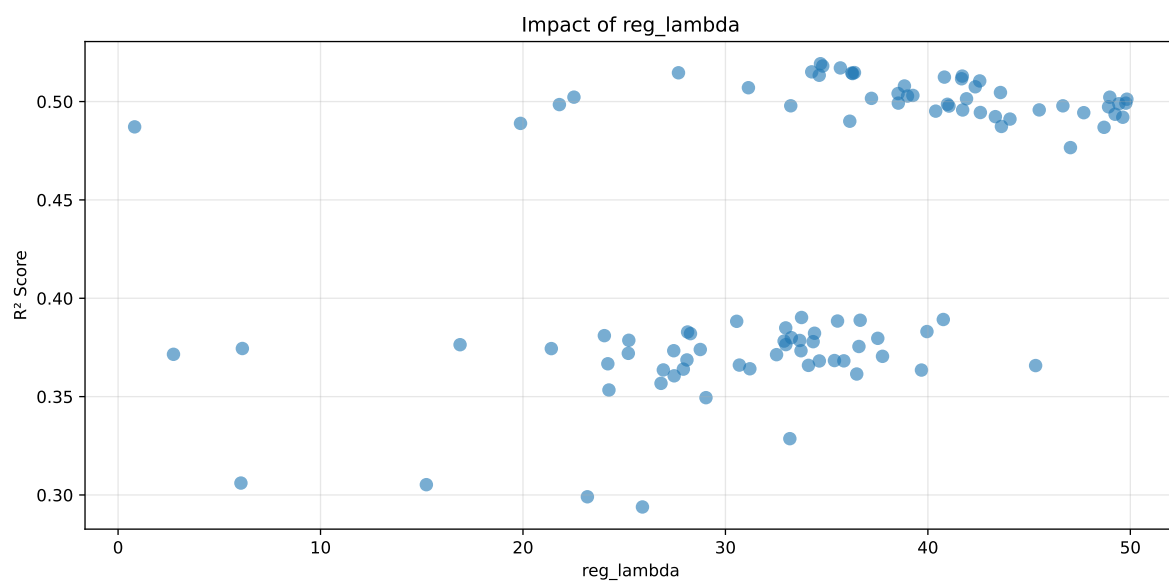
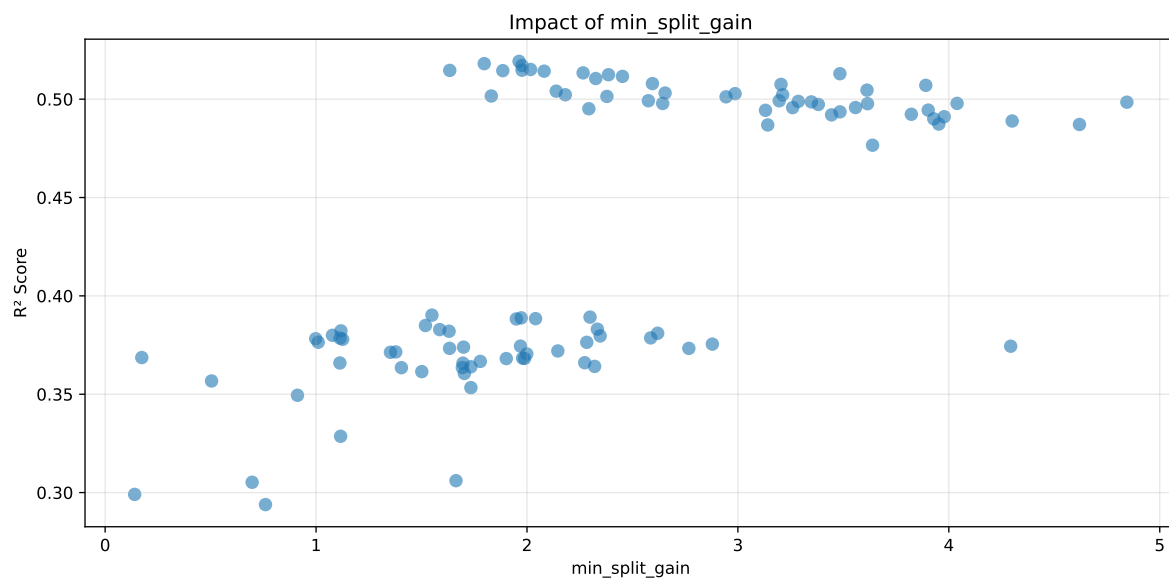
Len(group) : 4082, Len(training) : 2858, Len(test) : 1224

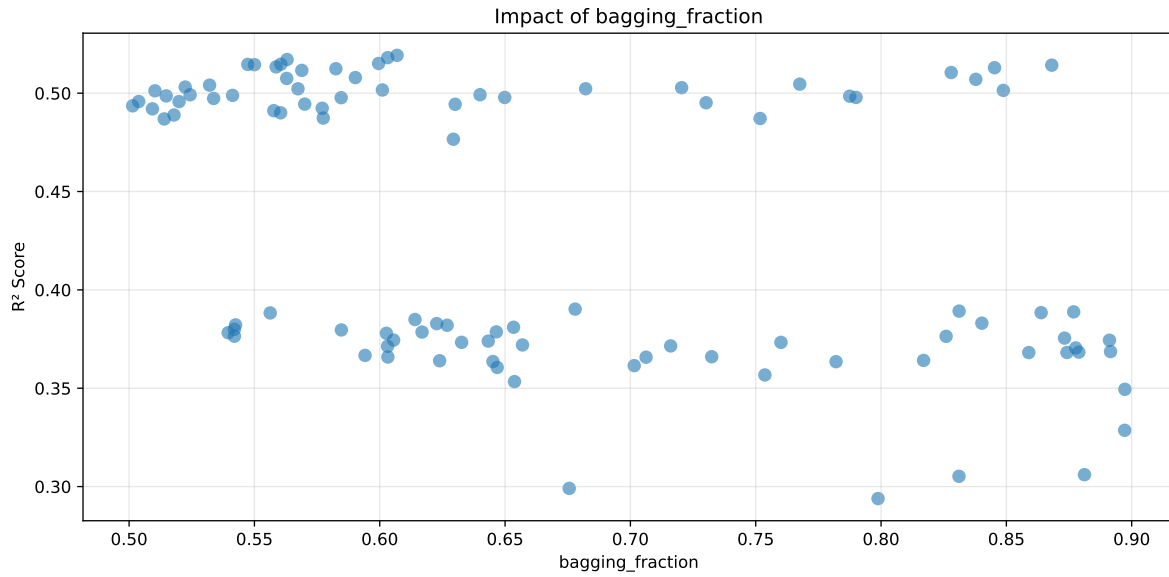
Distribution y_train vs y_test: y_train: mean=2.469, std=0.854, min=1.041, max=13.845
y_test: mean=2.472, std=0.811, min=1.068, max=7.277

===== TOP Importance FEATURES =====
feature importance r_NH3 191 r_umidity 181 r_CO2 162 r_temperature 84 r_THI 52



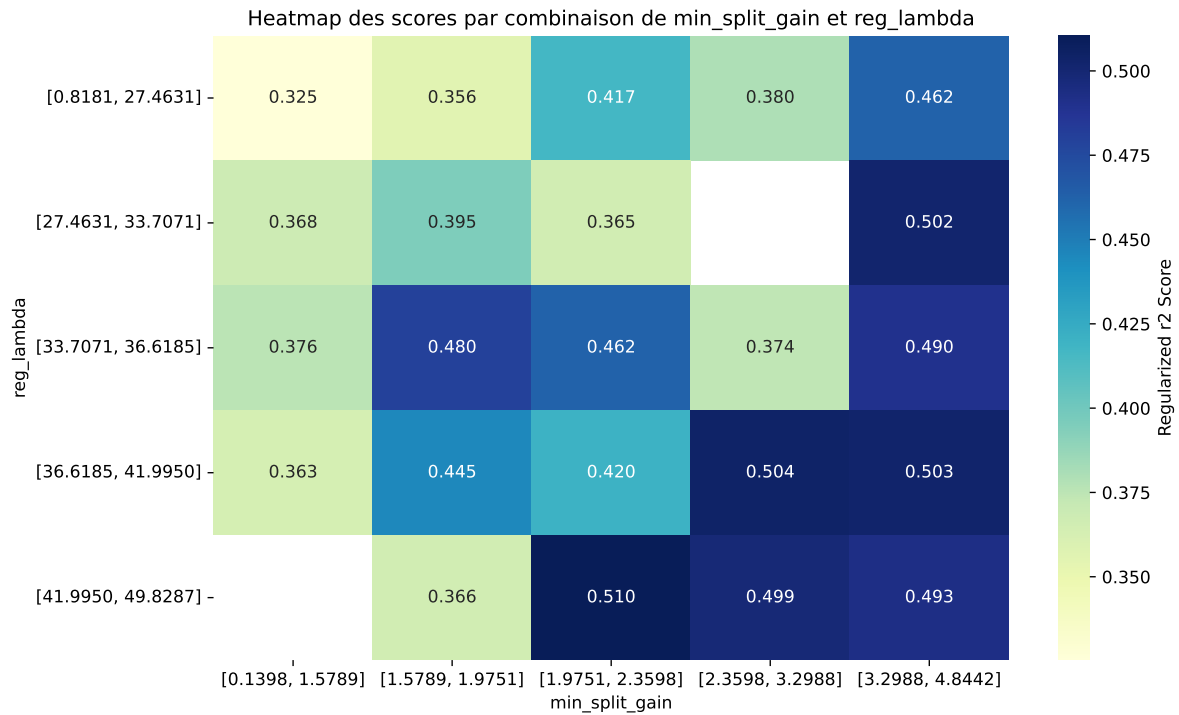
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_lambda', 'bagging_fraction']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.545	0.5192	0.5601	0.0603	0.5192	0.5845	0.0653	1.1258	1.0495
2	0.5432	0.5181	0.5596	0.065	0.5181	0.5883	0.0703	1.1356	1.0485
3	0.54	0.5171	0.5566	0.0617	0.5171	0.5774	0.0603	1.1166	1.0443
4	0.5393	0.5151	0.5561	0.063	0.5151	0.5815	0.0664	1.129	1.0471
5	0.538	0.5147	0.5544	0.0587	0.5147	0.5745	0.0598	1.1162	1.0451
6	0.5395	0.5146	0.5548	0.0608	0.5146	0.5783	0.0636	1.1237	1.0484
7	0.5375	0.5145	0.5537	0.0575	0.5145	0.5667	0.0522	1.1015	1.0447
8	0.5376	0.5142	0.5536	0.0605	0.5142	0.5762	0.0619	1.1204	1.0455
9	0.5383	0.5134	0.5539	0.0561	0.5134	0.5761	0.0627	1.1221	1.0485
10	0.537	0.5129	0.5525	0.0591	0.5129	0.5786	0.0657	1.1281	1.0469
---	---	---	---	---	---	---	---	---	---
100	0.6047	0.5529	0.5992	0.0609	0.2939	0.6168	0.0639	1.1155	1.0937

Hyperparameters:

Rank 1: {'reg__n_estimators': 600, 'reg__max_depth': 7, 'reg__learning_rate': 0.03238617131421312, 'reg__num_leaves': 44, 'reg__subsample': 0.5250844946152786, 'reg__colsample_bytree': 0.5849173658638209, 'reg__min_child_samples': 20, 'reg__reg_alpha': 3.0598747342229675, 'reg__reg_lambda': 34.694960223946744, 'reg__min_split_gain': 1.9625556071031778, 'reg__path_smooth': 0.4007860724098714, 'reg__min_child_weight': 6.305864609804129, 'reg__feature_fraction': 0.7977256686600315, 'reg__bagging_fraction': 0.6069085591317678, 'reg__bagging_freq': 3, 'reg__max_bin': 76, 'reg__min_data_per_group': 153}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.03199619407421765, 'reg__num_leaves': 35, 'reg__subsample': 0.5250116372681535, 'reg__colsample_bytree': 0.5894604201717135, 'reg__min_child_samples': 13, 'reg__reg_alpha': 2.9942974681867405, 'reg__reg_lambda': 34.80587162529053, 'reg__min_split_gain': 1.7972874083308943, 'reg__path_smooth': 0.2584258198955841, 'reg__min_child_weight': 5.575656125063763, 'reg__feature_fraction': 0.8013399962907591, 'reg__bagging_fraction': 0.6031748318984027, 'reg__bagging_freq': 3, 'reg__max_bin': 76, 'reg__min_data_per_group': 150}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.029773225812063292, 'reg__num_leaves': 37, 'reg__subsample': 0.5893334449655107, 'reg__colsample_bytree': 0.6165748775374238, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.468231691331332, 'reg__reg_lambda': 35.677327605274925, 'reg__min_split_gain': 1.9764970280835645, 'reg__path_smooth': 0.6778246042962937, 'reg__min_child_weight': 1.2761994442342859, 'reg__feature_fraction': 0.82365382724523, 'reg__bagging_fraction': 0.5630320245327589, 'reg__bagging_freq': 4, 'reg__max_bin': 75, 'reg__min_data_per_group': 172}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.031670318424207186, 'reg__num_leaves': 44, 'reg__subsample': 0.5268915010700966, 'reg__colsample_bytree': 0.5834834779722794, 'reg__min_child_samples': 13, 'reg__reg_alpha': 2.96017634872467, 'reg__reg_lambda': 34.26217656037211, 'reg__min_split_gain': 2.0168881018901113, 'reg__path_smooth': 0.6088184795511541, 'reg__min_child_weight': 6.08118223056024, 'reg__feature_fraction': 0.7928301635760612, 'reg__bagging_fraction': 0.5995503477320013, 'reg__bagging_freq': 3, 'reg__max_bin': 75, 'reg__min_data_per_group': 150}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.03948873100699714, 'reg__num_leaves': 47, 'reg__subsample': 0.5030714928330445, 'reg__colsample_bytree': 0.6187610802814123, 'reg__min_child_samples': 11, 'reg__reg_alpha': 2.3262934301529286, 'reg__reg_lambda': 36.37269482054358, 'reg__min_split_gain': 1.9767019338658836, 'reg__path_smooth': 0.37351976499633244, 'reg__min_child_weight': 8.957326004833087, 'reg__feature_fraction': 0.8316715000586129, 'reg__bagging_fraction': 0.5605319165775007, 'reg__bagging_freq': 5, 'reg__max_bin': 77, 'reg__min_data_per_group': 147}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.010457784585461033, 'reg__num_leaves': 50, 'reg__subsample': 0.6039525338816539, 'reg__colsample_bytree': 0.5412700618302536, 'reg__min_child_samples': 18, 'reg__reg_alpha': 4.33636288952674, 'reg__reg_lambda': 27.68041308906676, 'reg__min_split_gain': 1.6339006013300867, 'reg__path_smooth': 0.03402544999291102, 'reg__min_child_weight': 0.6539623810492632, 'reg__feature_fraction': 0.7641459792219919, 'reg__bagging_fraction': 0.5473545843726663, 'reg__bagging_freq': 2, 'reg__max_bin': 97, 'reg__min_data_per_group': 168}

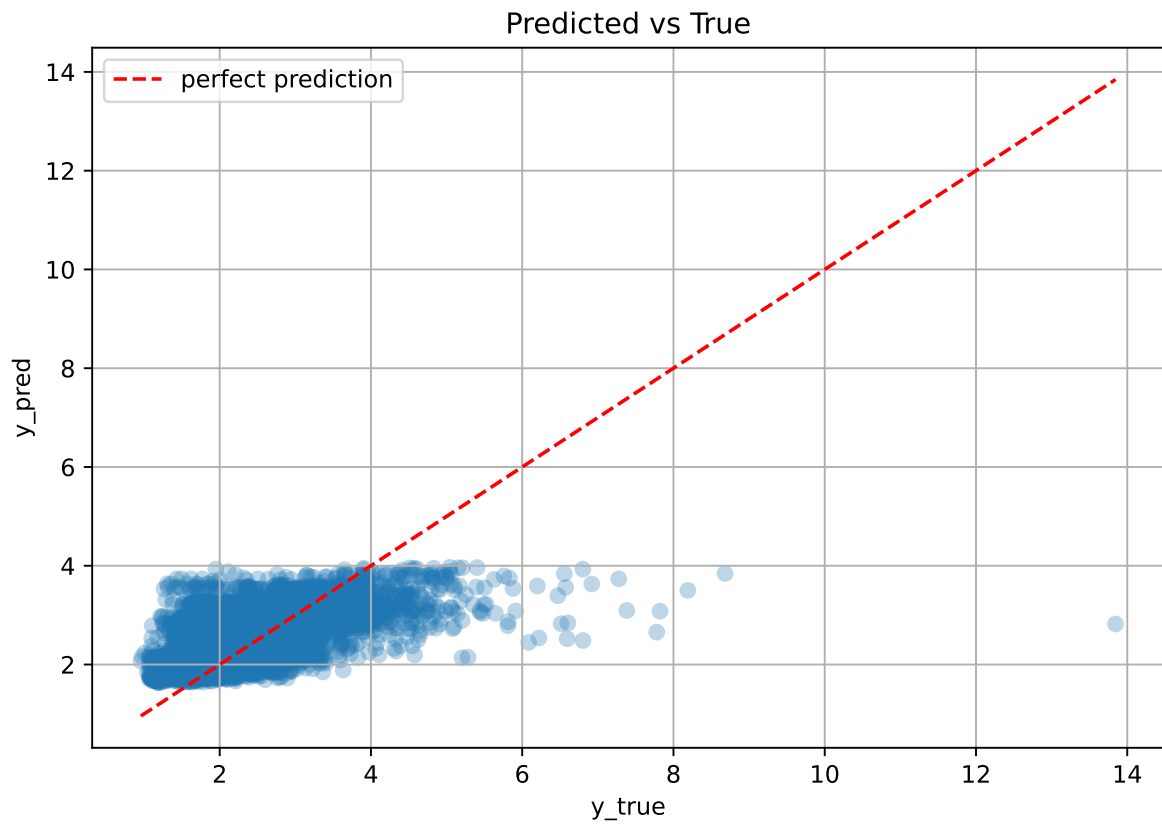
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.029854257585769647, 'reg__num_leaves': 36, 'reg__subsample': 0.5905944226508113, 'reg__colsample_bytree': 0.6281568997605268, 'reg__min_child_samples': 20, 'reg__reg_alpha': 3.092996696719793, 'reg__reg_lambda': 36.241627236247815, 'reg__min_split_gain': 1.8853630762145945, 'reg__path_smooth': 0.45174722491589536, 'reg__min_child_weight': 7.357201861302774, 'reg__feature_fraction': 0.8400989157407084, 'reg__bagging_fraction': 0.550023749023112, 'reg__bagging_freq': 4, 'reg__max_bin': 72, 'reg__min_data_per_group': 150}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.031404365550045, 'reg__num_leaves': 45, 'reg__subsample': 0.5280997727066827, 'reg__colsample_bytree': 0.6196971724274026, 'reg__min_child_samples': 20, 'reg__reg_alpha': 7.056454415114393, 'reg__reg_lambda': 36.28181487010845, 'reg__min_split_gain': 2.0813589206277867, 'reg__path_smooth': 0.9999140981399409, 'reg__min_child_weight': 9.585791465927244, 'reg__feature_fraction': 0.8486018716169587, 'reg__bagging_fraction': 0.868109841755961, 'reg__bagging_freq': 4, 'reg__max_bin': 75, 'reg__min_data_per_group': 143}

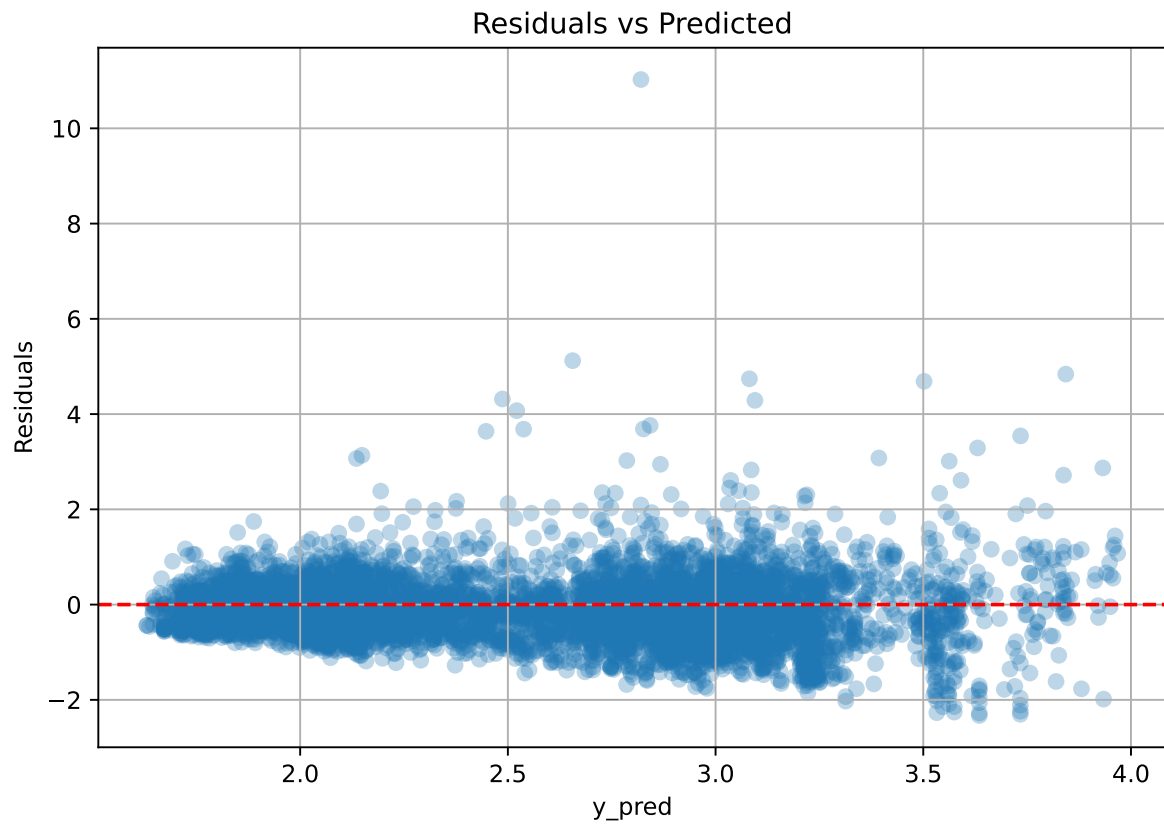
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.03954520756715224, 'reg__num_leaves': 39, 'reg__subsample': 0.5864488449757767, 'reg__colsample_bytree': 0.614872092227151, 'reg__min_child_samples': 10, 'reg__reg_alpha': 2.4547300917679005, 'reg__reg_lambda': 34.633195377208, 'reg__min_split_gain': 2.266090972118657, 'reg__path_smooth': 1.5381008809409558, 'reg__min_child_weight': 1.0373023045474297, 'reg__feature_fraction': 0.833562170941543, 'reg__bagging_fraction': 0.5586396837744202, 'reg__bagging_freq': 5, 'reg__max_bin': 82, 'reg__min_data_per_group': 111}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.028938917323256562, 'reg__num_leaves': 40, 'reg__subsample': 0.5517299124879063, 'reg__colsample_bytree': 0.6382047139523108, 'reg__min_child_samples': 15, 'reg__reg_alpha': 0.8770326446764715, 'reg__reg_lambda': 41.69907075461595, 'reg__min_split_gain': 3.4836054262297074, 'reg__path_smooth': 1.5331543489997568, 'reg__min_child_weight': 2.374016073437915, 'reg__feature_fraction': 0.8892667655607898, 'reg__bagging_fraction': 0.8452803131197103, 'reg__bagging_freq': 5, 'reg__max_bin': 128, 'reg__min_data_per_group': 95}

2.3.2 Scatter



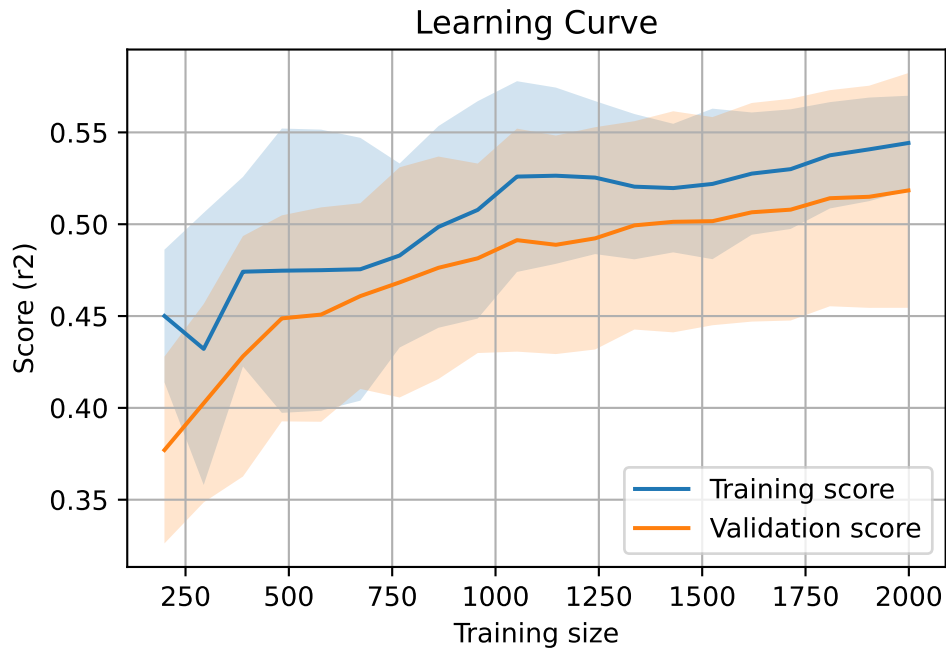
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 155

`Len(group)` : 4082, `Len(training)` : 2858,

`Len(training_real)` : 2001, `Len(test_of_training)` : 857, `Len(test)` : 1224

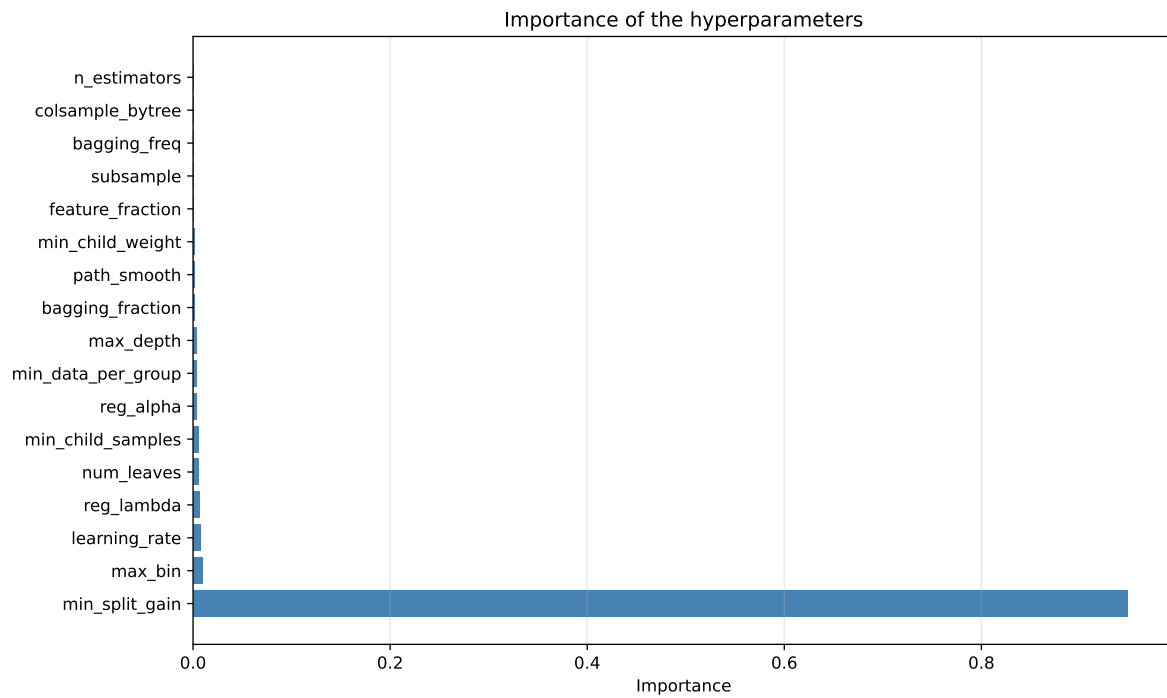
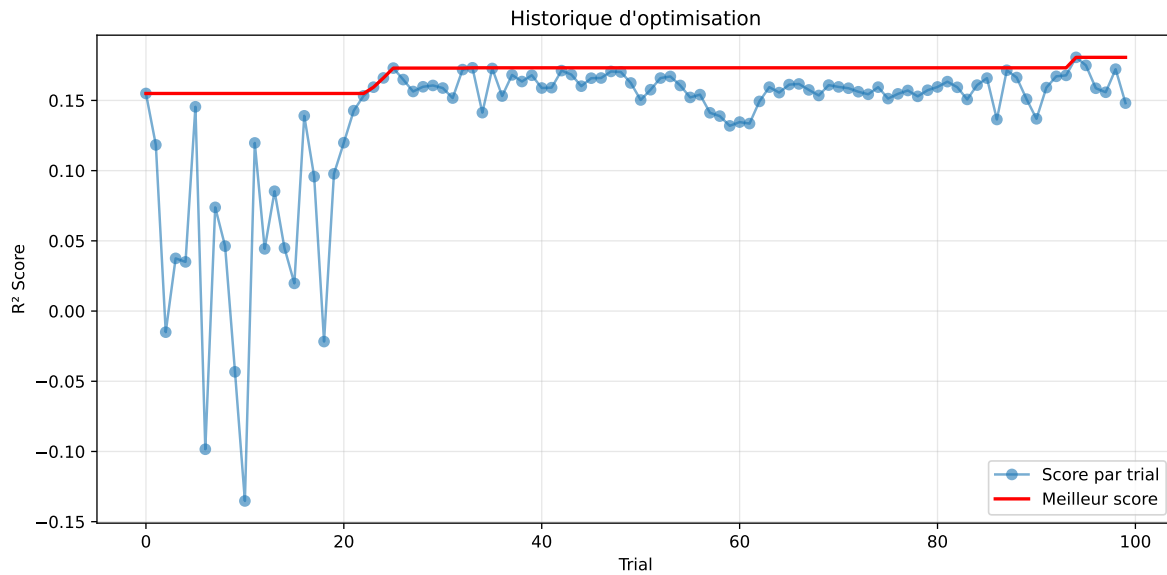


2.3.5 Gridsearch — Group: 270

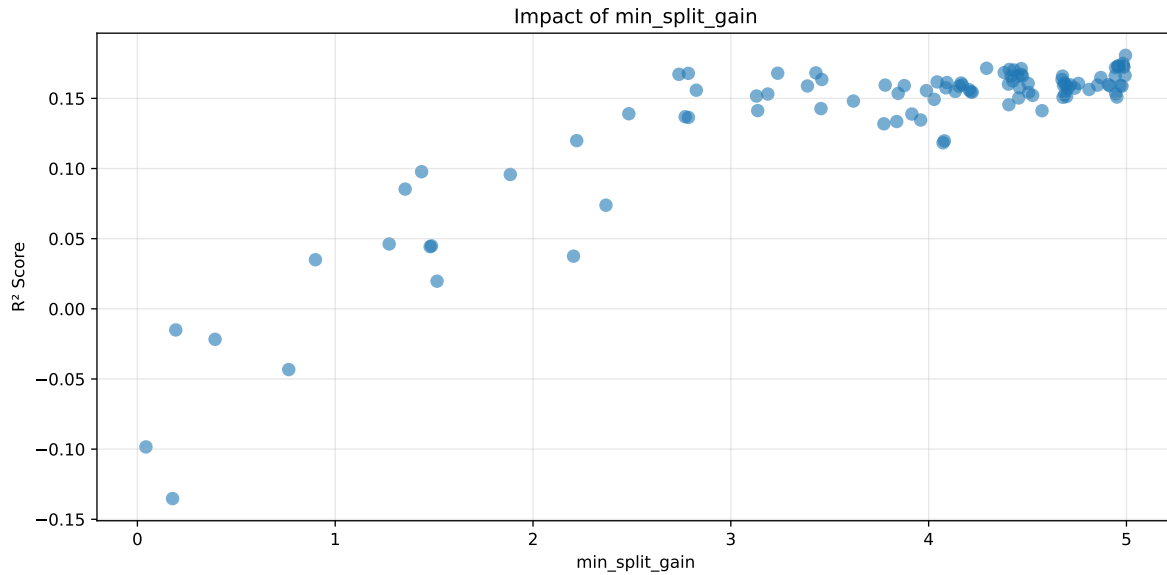
Len(group) : 4086, Len(training) : 2861, Len(test) : 1225

Distribution y_train vs y_test: y_train: mean=2.228, std=0.729, min=0.959, max=7.822
y_test: mean=2.232, std=0.719, min=0.973, max=6.804

===== TOP Importance FEATURES =====
feature importance r_umidity 185 r_THI 56 r_NH3 43 r_temperature 38 r_CO2 5



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3156	0.2931	0.3324	0.026	0.1807	0.3489	0.0558	1.1902	1.0767
2	0.3234	0.2987	0.3405	0.028	0.175	0.3469	0.0482	1.1613	1.0829
3	0.3336	0.3069	0.3497	0.0272	0.1732	0.3706	0.0637	1.2077	1.0871
4	0.3225	0.2976	0.3403	0.0257	0.173	0.362	0.0645	1.2166	1.0837
5	0.3578	0.327	0.3725	0.0292	0.1728	0.3929	0.0659	1.2015	1.0943
6	0.3365	0.3091	0.3512	0.0299	0.1723	0.3659	0.0568	1.1837	1.0886
7	0.3369	0.3094	0.353	0.0286	0.1719	0.3734	0.064	1.207	1.0888
8	0.3441	0.3154	0.3603	0.0291	0.1715	0.3846	0.0693	1.2196	1.0913
9	0.3218	0.2967	0.3389	0.0308	0.1712	0.3662	0.0695	1.2341	1.0846
10	0.3291	0.3027	0.3452	0.0256	0.1706	0.3786	0.0759	1.2508	1.0873
---	---	---	---	---	---	---	---	---	---
100	0.5488	0.4348	0.4937	0.0326	-0.1352	0.5269	0.0921	1.2118	1.2622

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.014600933810682346, 'reg__num_leaves': 44, 'reg__subsample': 0.5965820019938234, 'reg__colsample_bytree': 0.7656472974288048, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.621084853641234, 'reg__reg_lambda': 42.83788289213701, 'reg__min_split_gain': 4.995256862422404, 'reg__path_smooth': 2.02573202424976, 'reg__min_child_weight': 7.506036538626767, 'reg__feature_fraction': 0.8534032648695794, 'reg__bagging_fraction': 0.5596385071245311, 'reg__bagging_freq': 2, 'reg__max_bin': 65, 'reg__min_data_per_group': 187}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.014803453798783587, 'reg__num_leaves': 51, 'reg__subsample': 0.5965071509509232, 'reg__colsample_bytree': 0.7163146371700181, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.643099626878941, 'reg__reg_lambda': 43.79244194609013, 'reg__min_split_gain': 4.982844990843265, 'reg__path_smooth': 2.1386947668339436, 'reg__min_child_weight': 6.740329188790881, 'reg__feature_fraction': 0.7328963114712542, 'reg__bagging_fraction': 0.5661021800729938, 'reg__bagging_freq': 2, 'reg__max_bin': 63, 'reg__min_data_per_group': 188}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.013418710298476, 'reg__num_leaves': 34, 'reg__subsample': 0.6636908094610353, 'reg__colsample_bytree': 0.551008653298421, 'reg__min_child_samples': 27, 'reg__reg_alpha': 1.992115793312675, 'reg__reg_lambda': 48.85568326365275, 'reg__min_split_gain': 4.958201630409038, 'reg__path_smooth': 1.8657711963861925, 'reg__min_child_weight': 9.491149169364329, 'reg__feature_fraction': 0.7394971361716265, 'reg__bagging_fraction': 0.5020937682781849, 'reg__bagging_freq': 3, 'reg__max_bin': 65, 'reg__min_data_per_group': 200}

Rank 4: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.01035356718723225, 'reg__num_leaves': 33, 'reg__subsample': 0.8905324778180485, 'reg__colsample_bytree': 0.622861318400231, 'reg__min_child_samples': 25, 'reg__reg_alpha': 9.980140761480506, 'reg__reg_lambda': 49.605126227029075, 'reg__min_split_gain': 4.984489190424791, 'reg__path_smooth': 1.7950892674196073, 'reg__min_child_weight': 7.2192811354577895, 'reg__feature_fraction': 0.7727458808623971, 'reg__bagging_fraction': 0.5127750032871091, 'reg__bagging_freq': 1, 'reg__max_bin': 69, 'reg__min_data_per_group': 185}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.013654461226093923, 'reg__num_leaves': 47, 'reg__subsample': 0.6512537221979222, 'reg__colsample_bytree': 0.6355921318770088, 'reg__min_child_samples': 25, 'reg__reg_alpha': 1.436176411344345, 'reg__reg_lambda': 43.8578972275279, 'reg__min_split_gain': 4.9564999779802354, 'reg__path_smooth': 1.9113113869587857, 'reg__min_child_weight': 1.6529036874077718, 'reg__feature_fraction': 0.8087552509039372, 'reg__bagging_fraction': 0.5440949616784411, 'reg__bagging_freq': 1, 'reg__max_bin': 64, 'reg__min_data_per_group': 187}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.014629451193558766, 'reg__num_leaves': 44, 'reg__subsample': 0.712763835668989, 'reg__colsample_bytree': 0.6841456062407446, 'reg__min_child_samples': 27, 'reg__reg_alpha': 1.8730297097591748, 'reg__reg_lambda': 39.63219427750592, 'reg__min_split_gain': 4.987281127166098, 'reg__path_smooth': 2.124182809677331, 'reg__min_child_weight': 7.015522993655984, 'reg__feature_fraction': 0.727618300052628, 'reg__bagging_fraction': 0.5011097576626637, 'reg__bagging_freq': 2, 'reg__max_bin': 68, 'reg__min_data_per_group': 187}

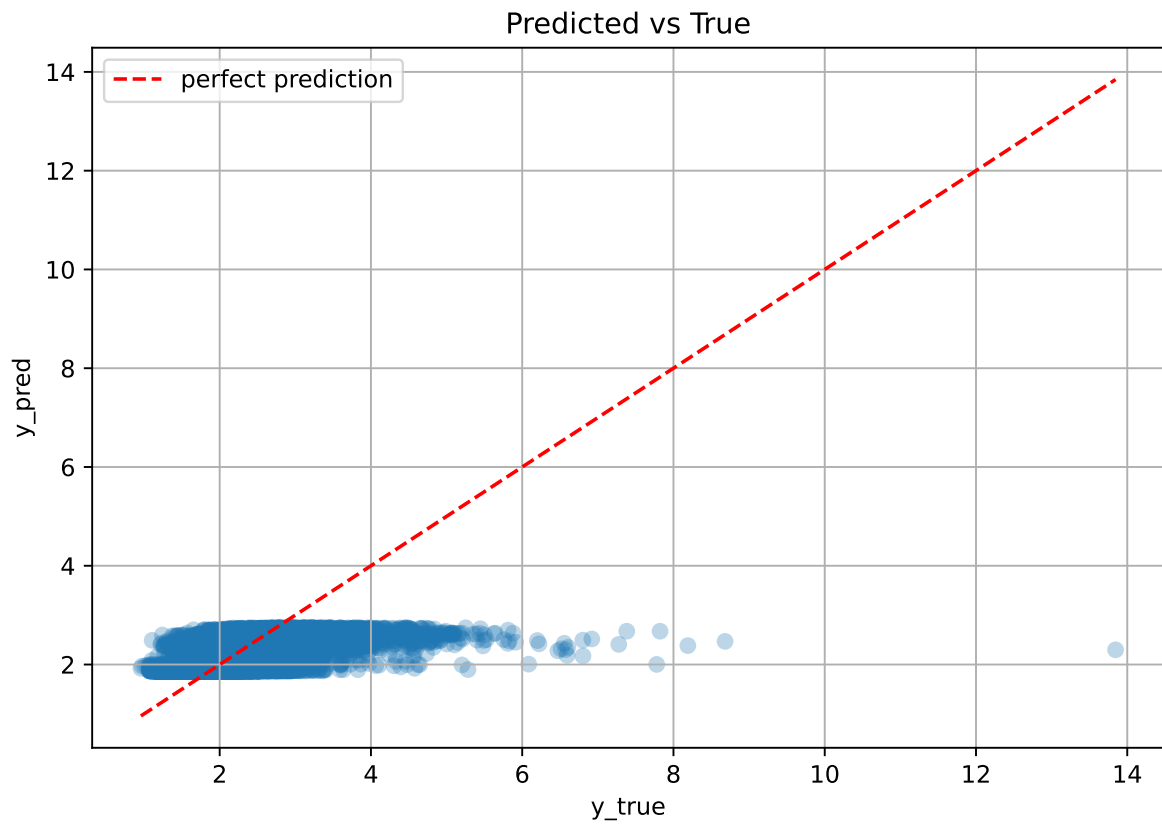
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.014286366082176677, 'reg__num_leaves': 33, 'reg__subsample': 0.648954285519018, 'reg__colsample_bytree': 0.5504316250934798, 'reg__min_child_samples': 28, 'reg__reg_alpha': 2.8380027166754886, 'reg__reg_lambda': 48.746077380901966, 'reg__min_split_gain': 4.94489089207435, 'reg__path_smooth': 2.2656853839030124, 'reg__min_child_weight': 9.137455452337628, 'reg__feature_fraction': 0.5163822944145987, 'reg__bagging_fraction': 0.5148851804327551, 'reg__bagging_freq': 1, 'reg__max_bin': 65, 'reg__min_data_per_group': 196}

Rank 8: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.011153153091985417, 'reg__num_leaves': 43, 'reg__subsample': 0.5938444873839082, 'reg__colsample_bytree': 0.5440207140197801, 'reg__min_child_samples': 27, 'reg__reg_alpha': 9.530983917738949, 'reg__reg_lambda': 44.39641771672043, 'reg__min_split_gain': 4.292993064147962, 'reg__path_smooth': 2.1827728283384453, 'reg__min_child_weight': 4.830106682676117, 'reg__feature_fraction': 0.7399731567226039, 'reg__bagging_fraction': 0.6285167118710537, 'reg__bagging_freq': 1, 'reg__max_bin': 64, 'reg__min_data_per_group': 188}

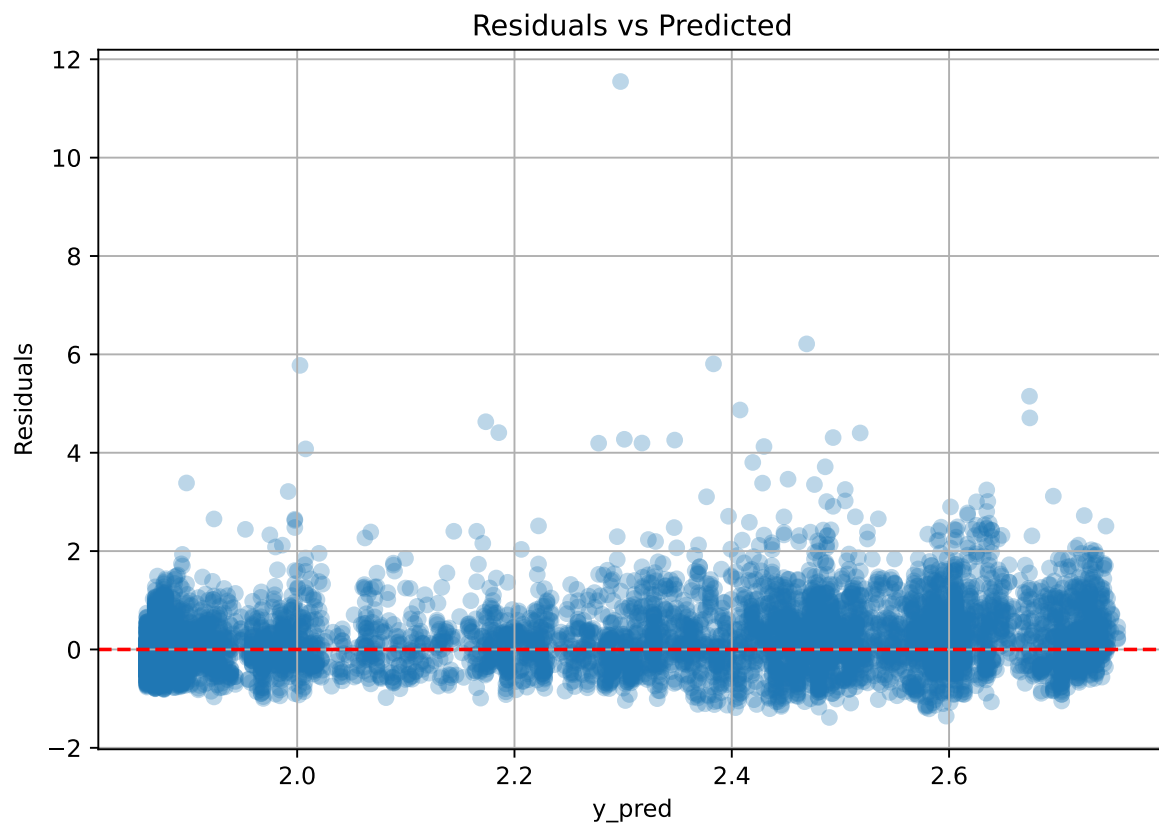
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 3, 'reg__learning_rate': 0.014626856926277725, 'reg__num_leaves': 26, 'reg__subsample': 0.6759155763381535, 'reg__colsample_bytree': 0.6747959914095845, 'reg__min_child_samples': 31, 'reg__reg_alpha': 9.86432491856078, 'reg__reg_lambda': 41.23623817940033, 'reg__min_split_gain': 4.468030249601915, 'reg__path_smooth': 2.301466052635945, 'reg__min_child_weight': 1.7827448641400498, 'reg__feature_fraction': 0.8009322962357331, 'reg__bagging_fraction': 0.550466068948473, 'reg__bagging_freq': 2, 'reg__max_bin': 94, 'reg__min_data_per_group': 184}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.016765177044811393, 'reg__num_leaves': 27, 'reg__subsample': 0.6140227892187679, 'reg__colsample_bytree': 0.5159372338766064, 'reg__min_child_samples': 31, 'reg__reg_alpha': 9.994826523659562, 'reg__reg_lambda': 38.27454607796822, 'reg__min_split_gain': 4.40792038152255, 'reg__path_smooth': 2.320895944485988, 'reg__min_child_weight': 0.4799958252734622, 'reg__feature_fraction': 0.7167543153546152, 'reg__bagging_fraction': 0.6374568472785516, 'reg__bagging_freq': 4, 'reg__max_bin': 95, 'reg__min_data_per_group': 187}

2.3.6 Scatter



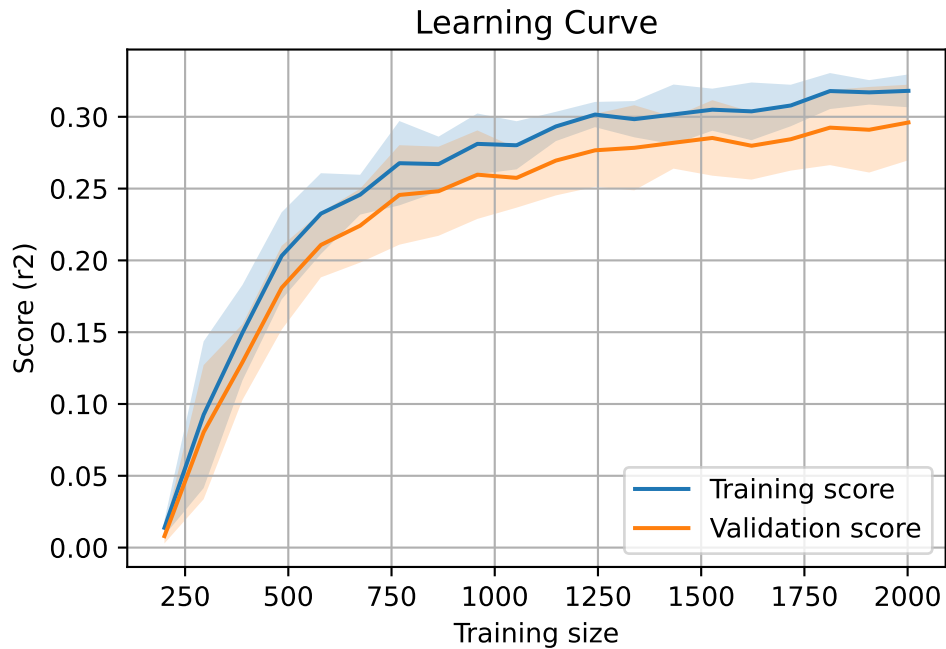
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 270

`Len(group)` : 4086, `Len(training)` : 2861,

`Len(training_real)` : 2003, `Len(test_of_training)` : 858, `Len(test)` : 1225



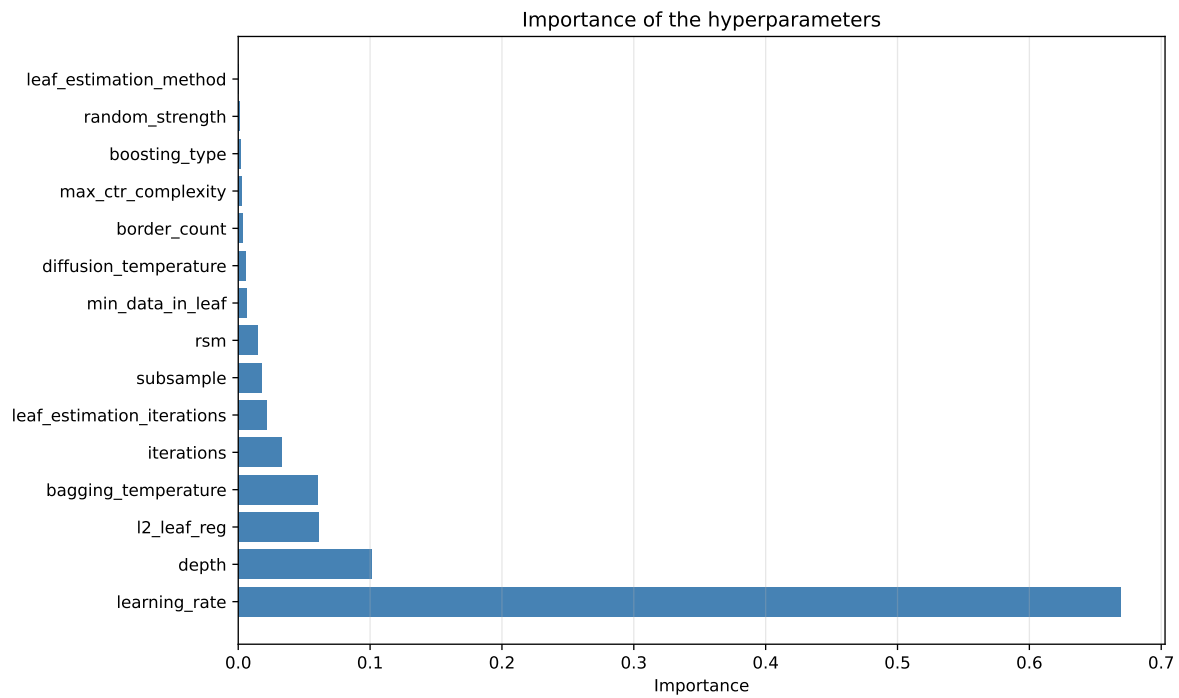
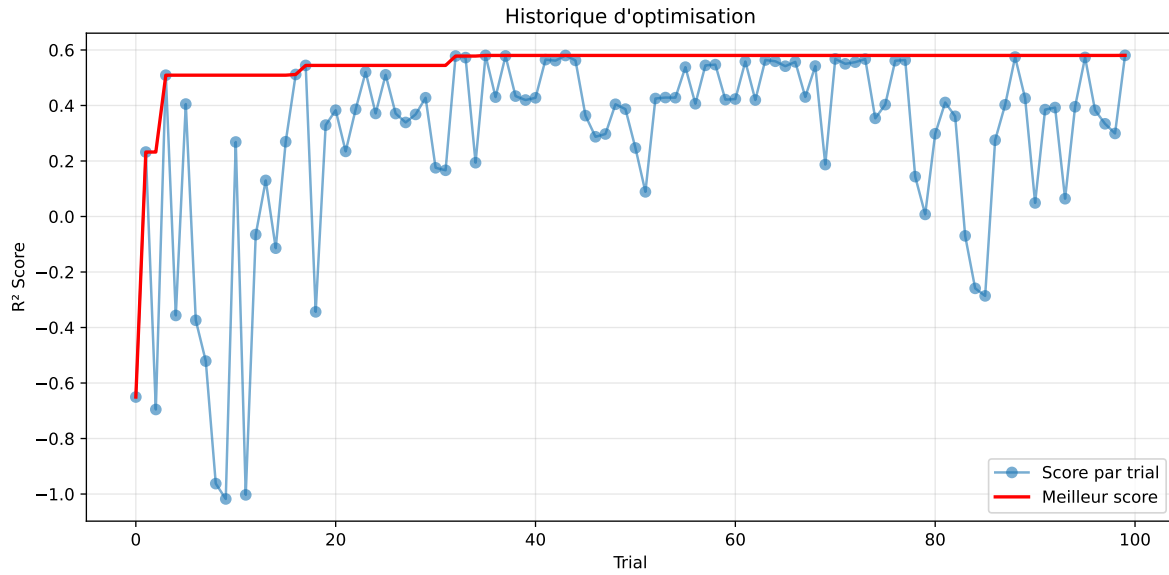
2.4 Model : CatBoost

2.4.1 Gridsearch — Group: 155

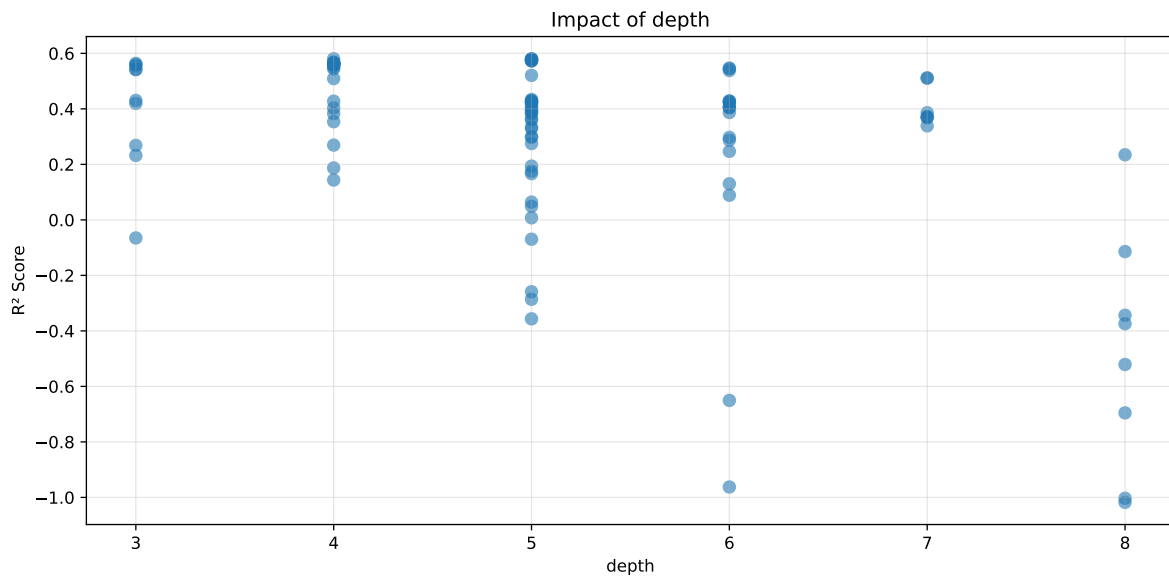
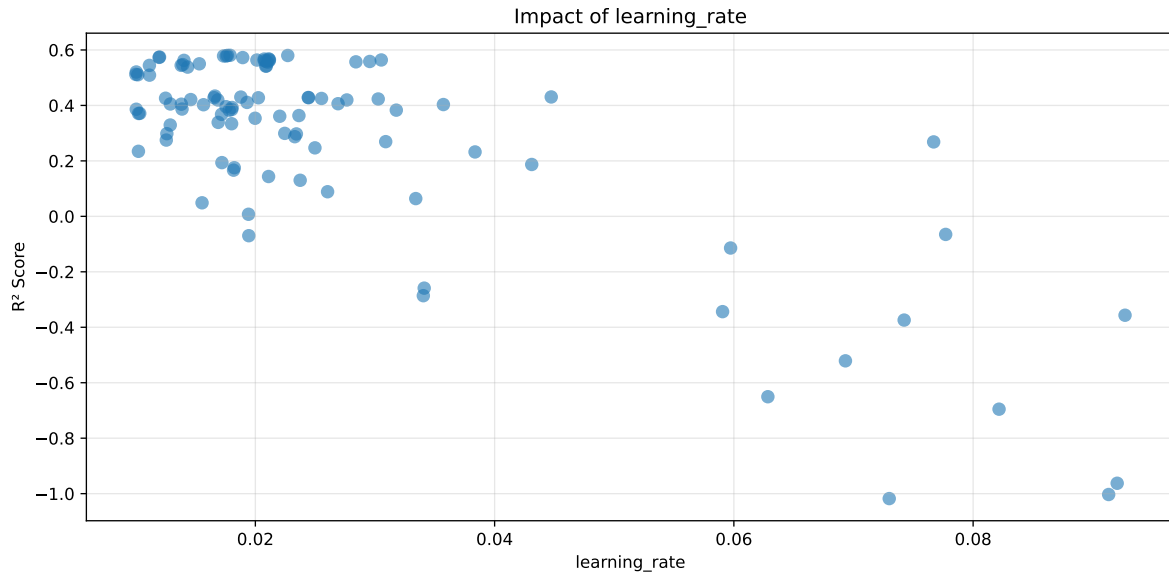
Len(group) : 4082, Len(training) : 2858, Len(test) : 1224

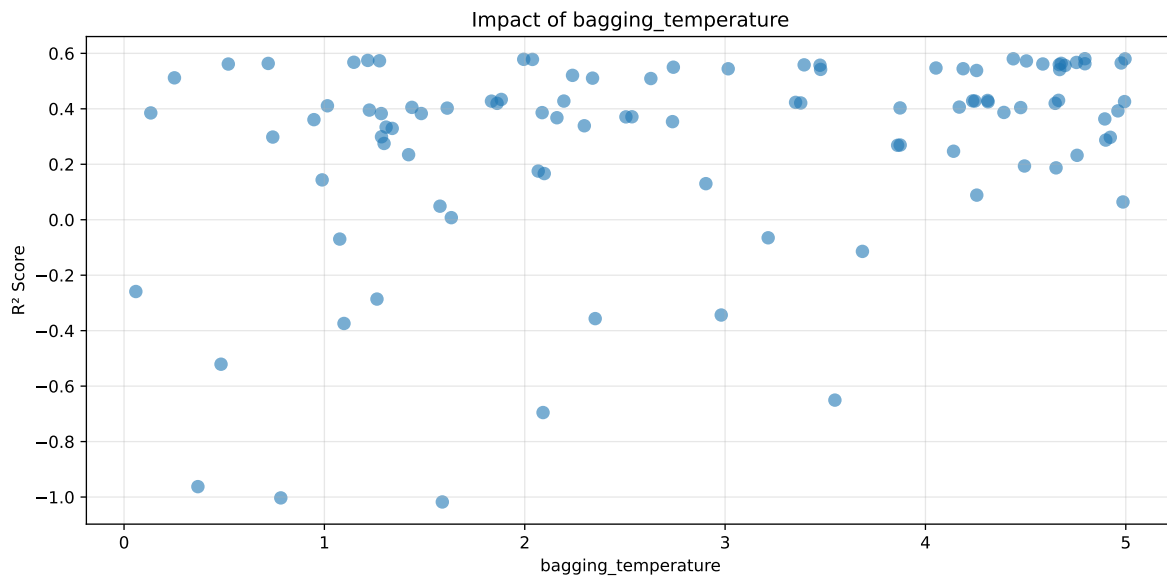
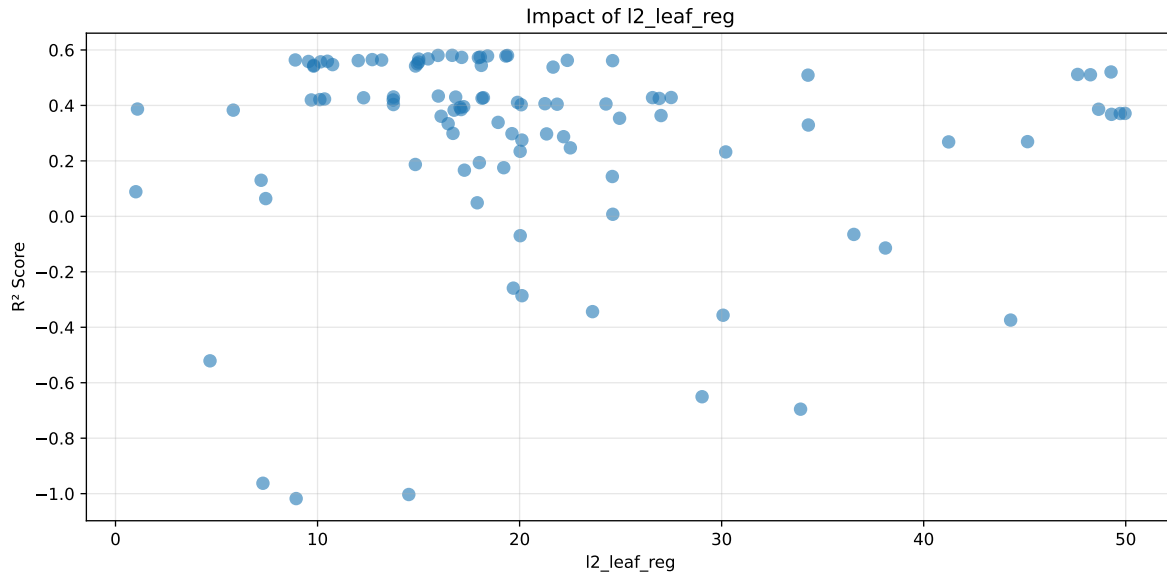
Distribution y_train vs y_test: y_train: mean=2.469, std=0.854, min=1.041, max=13.845
y_test: mean=2.472, std=0.811, min=1.068, max=7.277

===== TOP Importance FEATURES =====
feature importance r_umidity 37.520078 r_NH3 31.156340 r_CO2 12.927097 r_temperature
9.513427 r_THI 8.883057



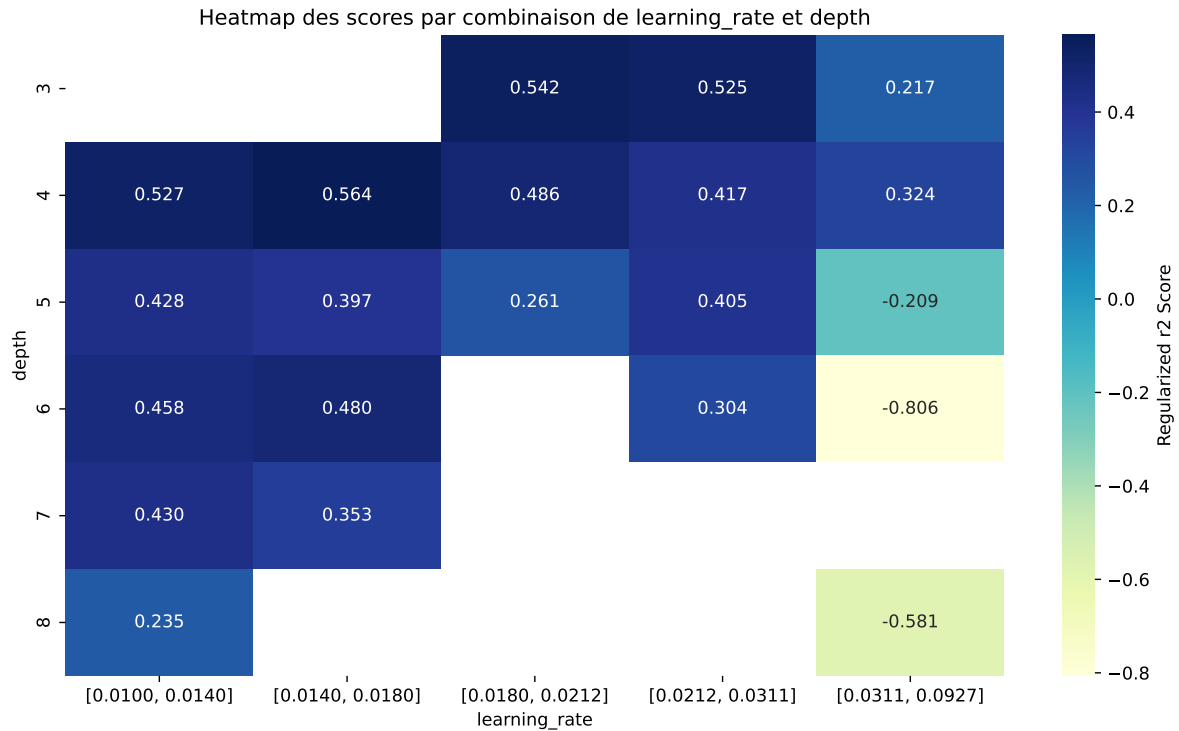
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'l2_leaf_reg', 'bagging_temperature']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6085	0.5806	0.6227	0.0593	0.5806	0.6297	0.0491	1.0845	1.048
2	0.6086	0.5803	0.6238	0.0595	0.5803	0.6347	0.0543	1.0936	1.0487
3	0.6082	0.5801	0.6267	0.0595	0.5801	0.633	0.0529	1.0912	1.0485
4	0.6064	0.5784	0.6231	0.0596	0.5784	0.6337	0.0552	1.0955	1.0484
5	0.6068	0.5781	0.6248	0.0622	0.5781	0.6364	0.0583	1.1008	1.0497
6	0.6024	0.5746	0.6179	0.0588	0.5746	0.6239	0.0493	1.0859	1.0484
7	0.6012	0.5732	0.6175	0.0588	0.5732	0.627	0.0538	1.0938	1.0489
8	0.5983	0.5725	0.6183	0.0593	0.5725	0.6236	0.051	1.0892	1.045
9	0.5913	0.5679	0.61	0.0589	0.5679	0.6161	0.0482	1.085	1.0413
10	0.5907	0.5672	0.6101	0.0596	0.5672	0.6151	0.0479	1.0844	1.0413
---	---	---	---	---	---	---	---	---	---
100	0.9544	0.6257	0.6772	0.0564	-1.0178	0.712	0.0863	1.1379	1.5253

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 4, 'reg__learning_rate': 0.017886743193372055, 'reg__l2_leaf_reg': 16.66317704229055, 'reg__bagging_temperature': 4.79569184451132, 'reg__random_strength': 2.348634069453306, 'reg__subsample': 0.796202254854981, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8036971098213525, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4850.878430024187}

Rank 2: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.017666477020505316, 'reg__l2_leaf_reg': 15.962725492550259, 'reg__bagging_temperature': 4.438354225341714, 'reg__random_strength': 1.0106688670842074, 'reg__subsample': 0.7147902769855317, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8160757935471297, 'reg__border_count': 94, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3253.460545169117}

Rank 3: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.02270875681561685, 'reg__l2_leaf_reg': 19.401609667941887, 'reg__bagging_temperature': 4.996176941329369, 'reg__random_strength': 2.6353069753937577, 'reg__subsample': 0.7237106876806507, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.849711908281758, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 7210.243687202916}

Rank 4: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.017341340707880015, 'reg__l2_leaf_reg': 18.41422405471026, 'reg__bagging_temperature': 1.9948503214080031, 'reg__random_strength': 0.9920073465999544, 'reg__subsample': 0.8300435323876981, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8162892583789925, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 5789.381046939919}

Rank 5: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.017561793727617087, 'reg__l2_leaf_reg': 19.321516274192167, 'reg__bagging_temperature': 2.0384555463703724, 'reg__random_strength': 0.9209231011368321, 'reg__subsample': 0.724366529210198, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8114330815219476, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6189.327918911247}

Rank 6: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.01198994039938566, 'reg__l2_leaf_reg': 18.055791578888915, 'reg__bagging_temperature': 1.2165837455434212, 'reg__random_strength': 2.2711954928721, 'reg__subsample': 0.738927315797445, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.8676750487888514, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4690.417263268342}

Rank 7: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.011951615768400134, 'reg__l2_leaf_reg': 17.132781754019284, 'reg__bagging_temperature': 1.2750199372102242, 'reg__random_strength': 2.187412425581283, 'reg__subsample': 0.7381772265656833,

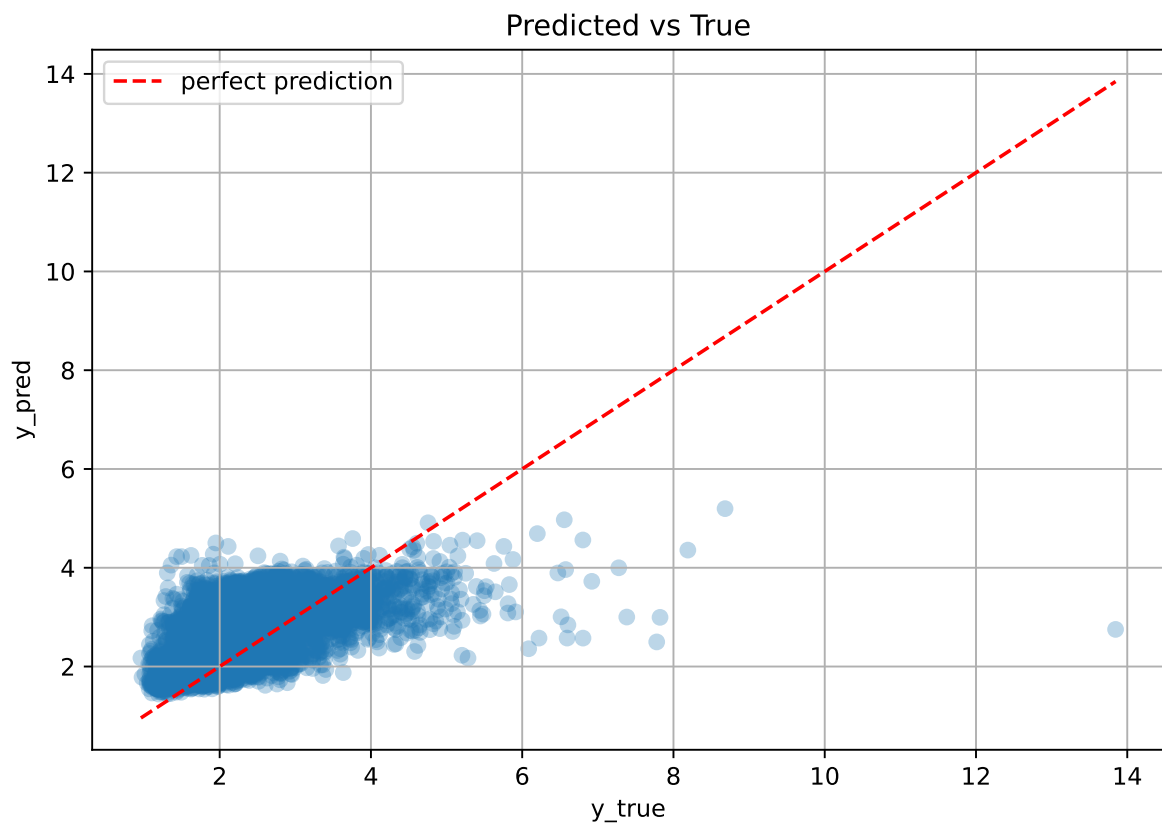
'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.873673464050485, 'reg__border_count': 89, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4696.547674879172}

Rank 8: {'reg__iterations': 300, 'reg__depth': 5, 'reg__learning_rate': 0.01894221955416465, 'reg__l2_leaf_reg': 17.96171909041106, 'reg__bagging_temperature': 4.503967541479685, 'reg__random_strength': 2.7305920920144904, 'reg__subsample': 0.7073037726711584, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8494879241877744, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 6219.544819570039}

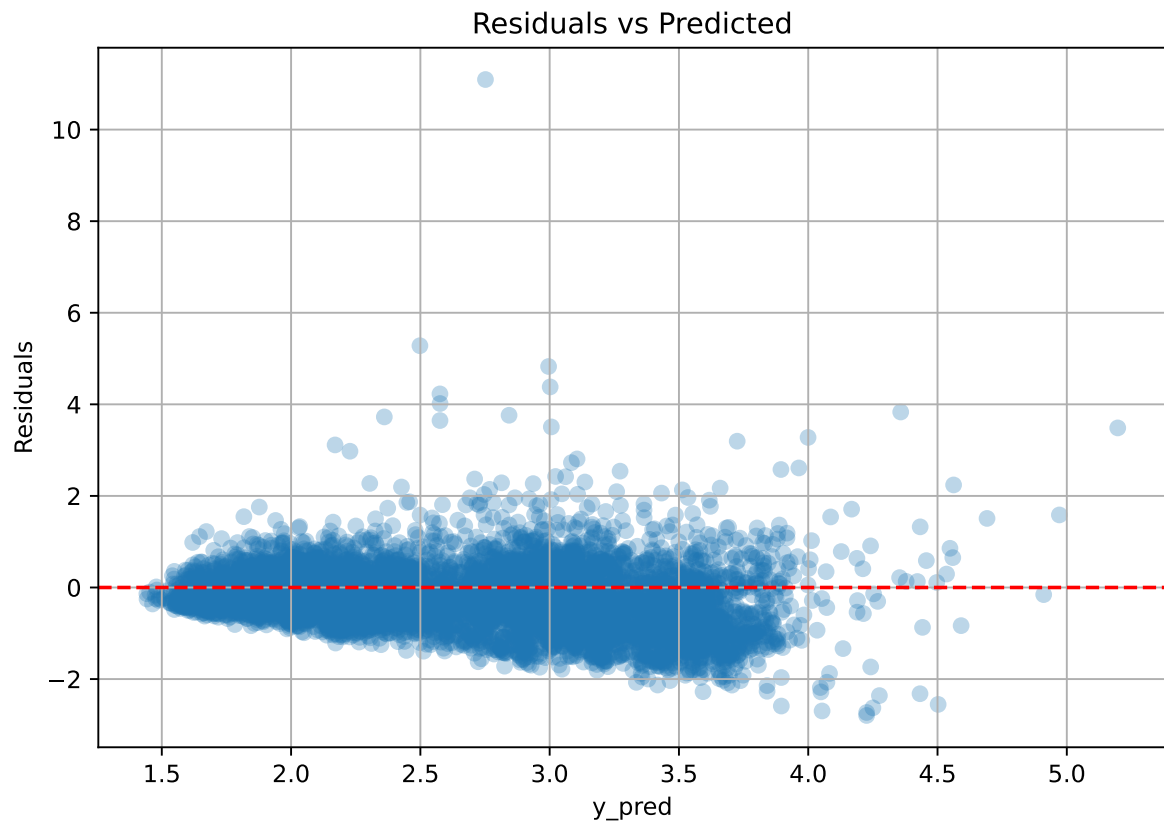
Rank 9: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.021114393414430378, 'reg__l2_leaf_reg': 15.470578386863009, 'reg__bagging_temperature': 1.1470020806863335, 'reg__random_strength': 3.1957719195452485, 'reg__subsample': 0.6960402903910962, 'reg__min_data_in_leaf': 50, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7914443260825595, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4587.720044189825}

Rank 10: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.02074018580048575, 'reg__l2_leaf_reg': 15.005574821297996, 'reg__bagging_temperature': 4.753644020862542, 'reg__random_strength': 3.009455456089274, 'reg__subsample': 0.6931544587042123, 'reg__min_data_in_leaf': 37, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8420323439236787, 'reg__border_count': 97, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 4607.980426877539}

2.4.2 Scatter



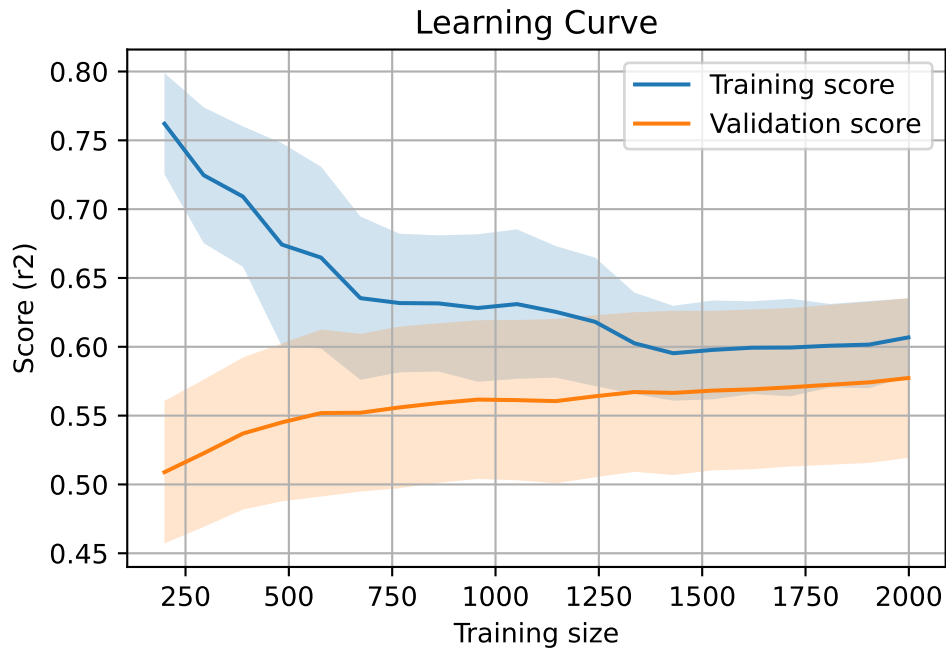
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 155

`Len(group)` : 4082, `Len(training)` : 2858,

`Len(training_real)` : 2001, `Len(test_of_training)` : 857, `Len(test)` : 1224

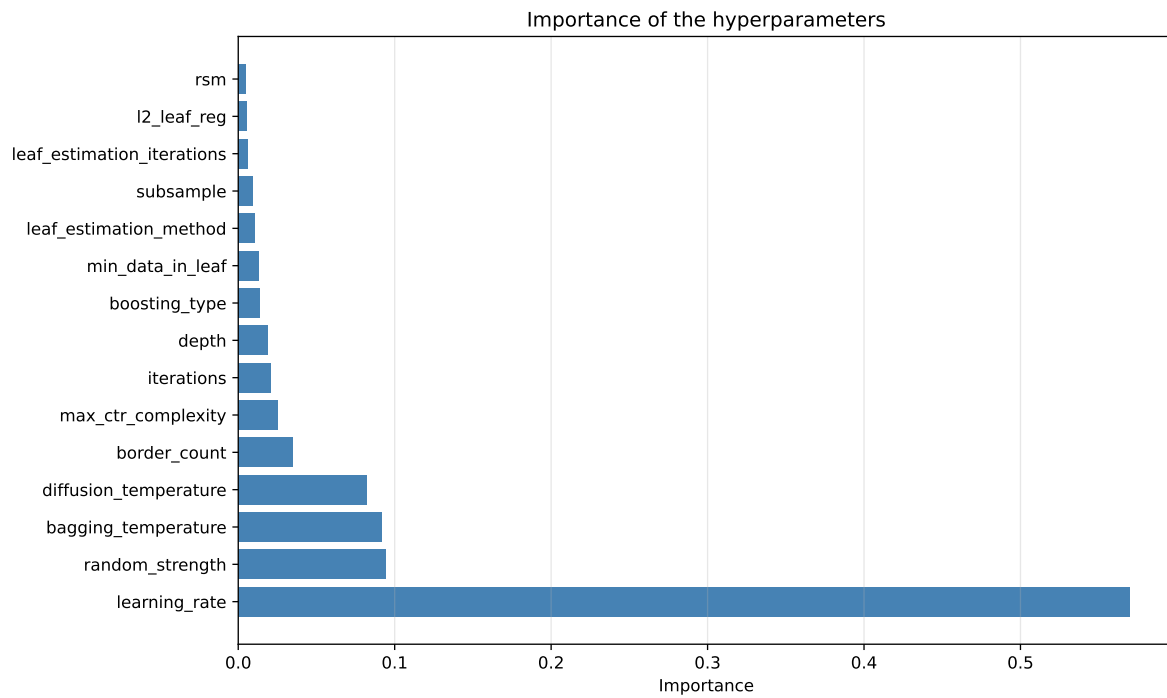
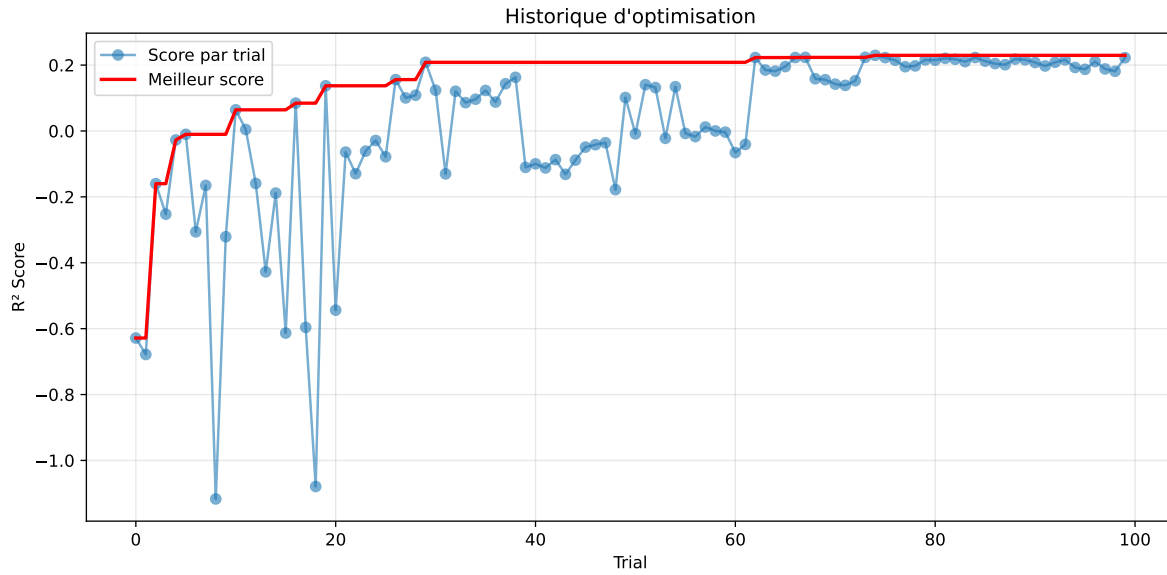


2.4.5 Gridsearch — Group: 270

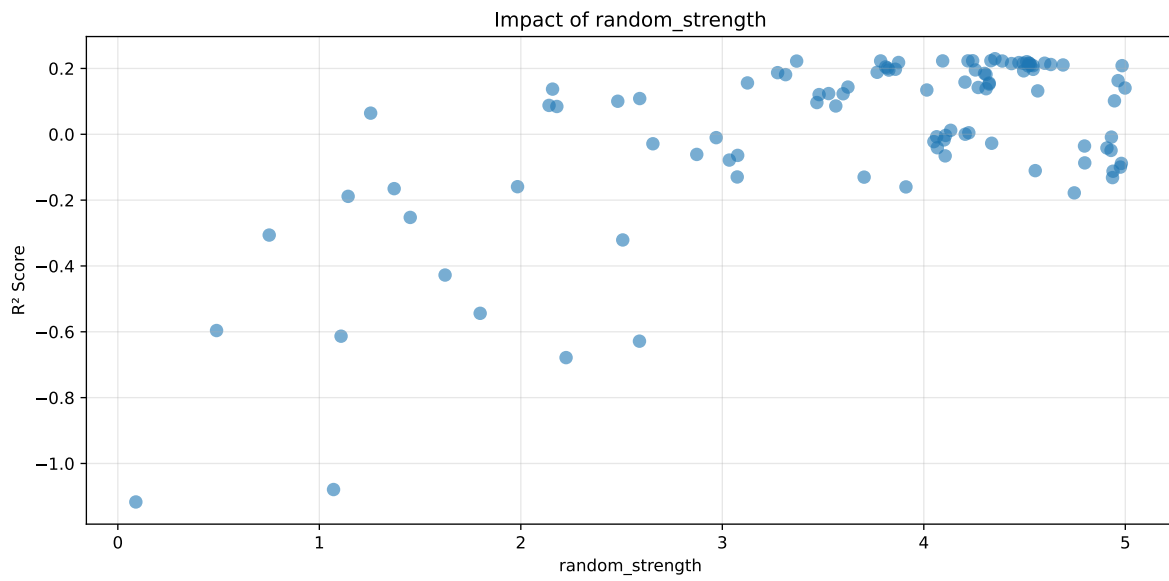
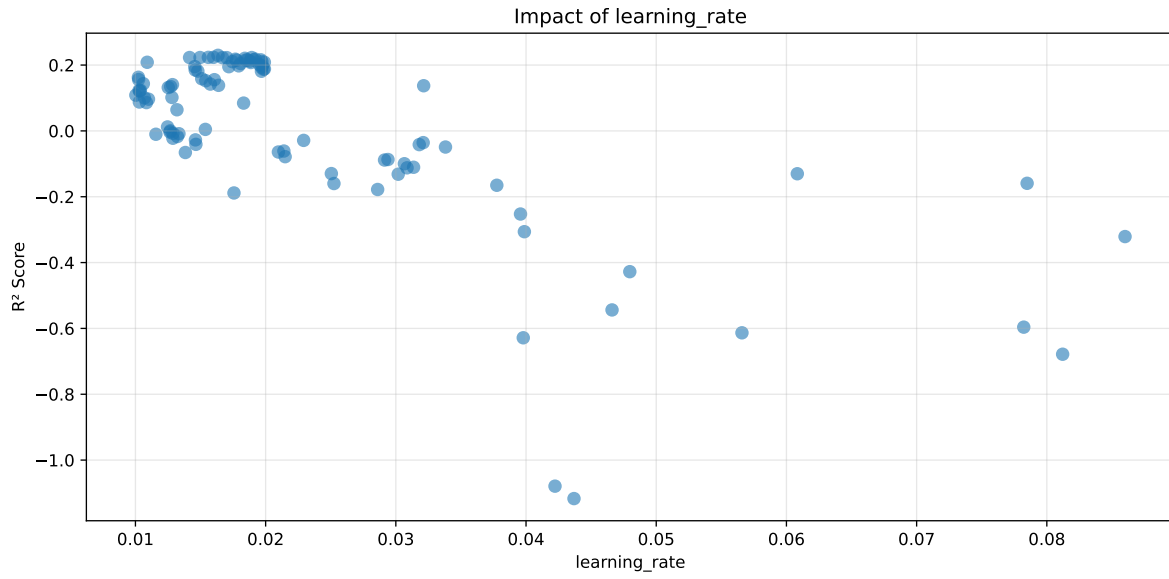
Len(group) : 4086, Len(training) : 2861, Len(test) : 1225

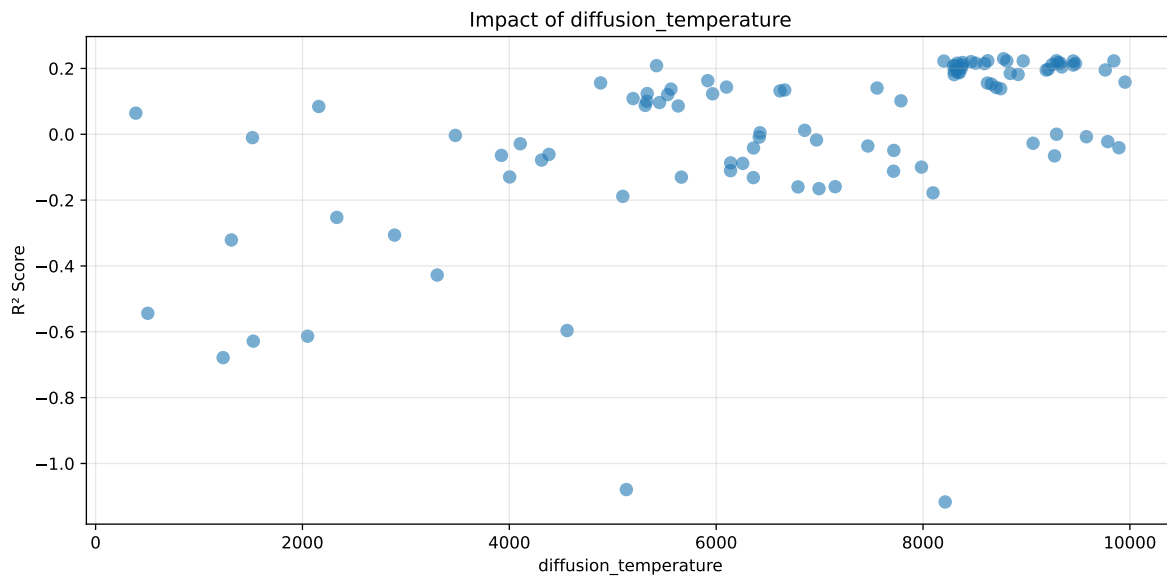
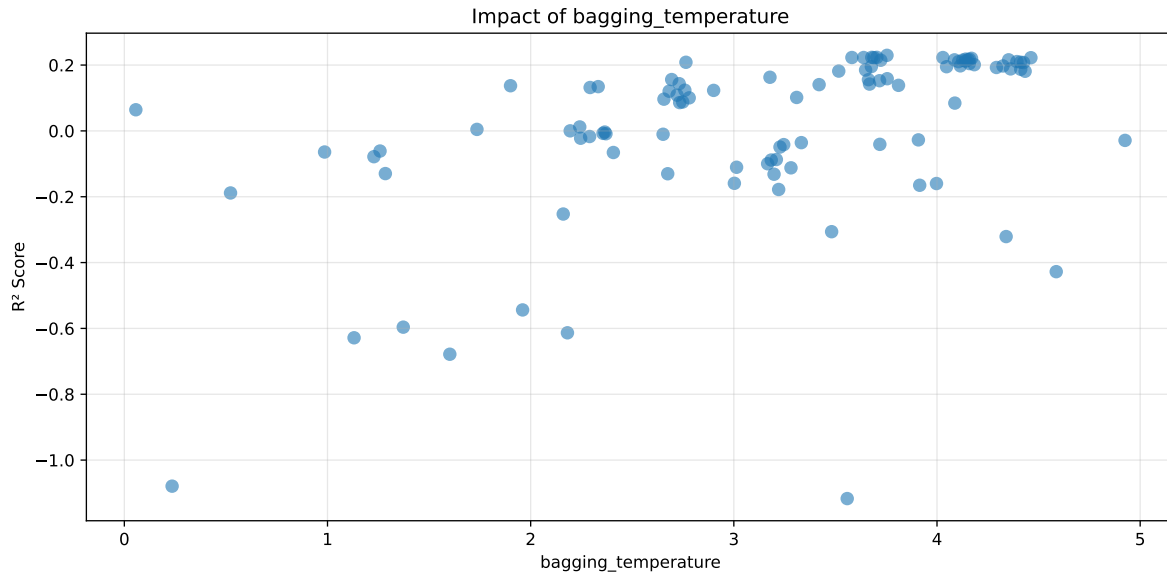
Distribution y_train vs y_test: y_train: mean=2.228, std=0.729, min=0.959, max=7.822
y_test: mean=2.232, std=0.719, min=0.973, max=6.804

===== TOP Importance FEATURES =====
feature importance r_umidity 62.690919 r_THI 12.518140 r_NH3 9.981507 r_temperature
7.604716 r_CO2 7.204719



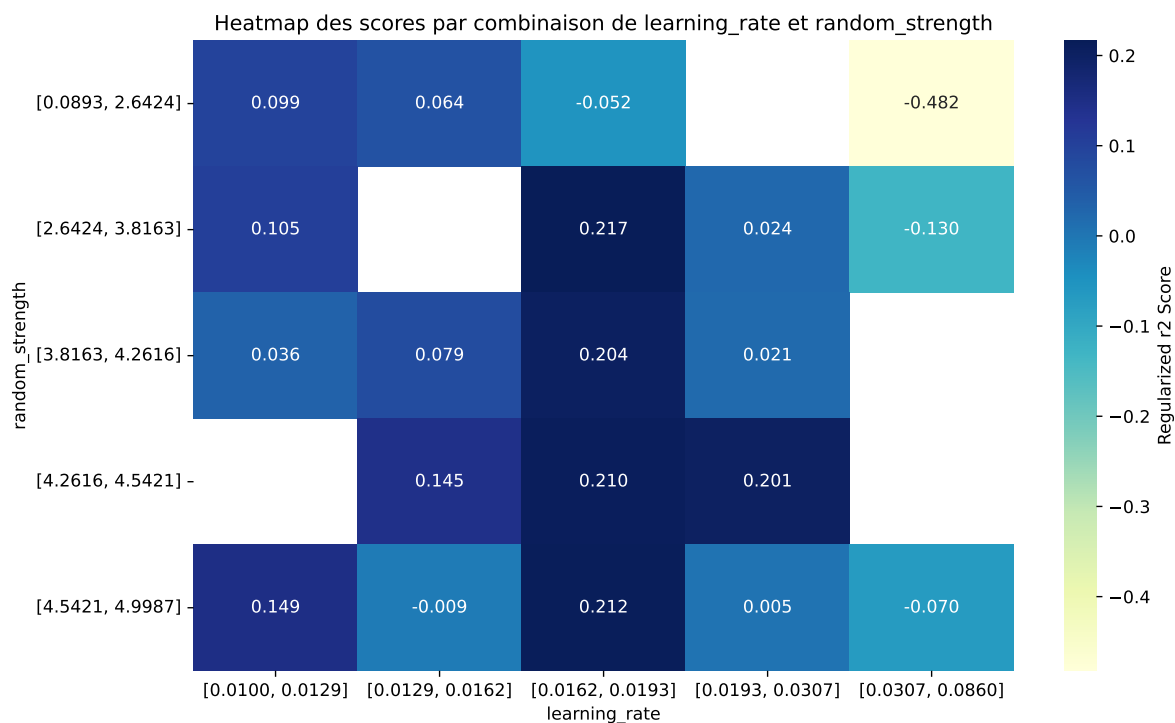
Paramètres avec >5% d'importance : ['learning_rate', 'random_strength', 'bagging_temperature', 'diffusion_temperature']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3846	0.3587	0.4066	0.029	0.2296	0.4061	0.0473	1.1319	1.072
2	0.3766	0.3511	0.3982	0.0288	0.2235	0.3988	0.0477	1.1359	1.0727
3	0.3749	0.3497	0.3964	0.0269	0.2234	0.4034	0.0538	1.1538	1.0722
4	0.3733	0.3482	0.3937	0.0256	0.2229	0.3976	0.0494	1.1418	1.072
5	0.3745	0.3492	0.3963	0.028	0.2229	0.3988	0.0495	1.1418	1.0724
6	0.435	0.3997	0.4475	0.031	0.2228	0.4502	0.0505	1.1265	1.0885
7	0.3823	0.3557	0.4043	0.0286	0.2225	0.4083	0.0526	1.148	1.0749
8	0.3931	0.3646	0.415	0.0277	0.2223	0.4141	0.0495	1.1356	1.078
9	0.4275	0.393	0.4429	0.0311	0.2205	0.4421	0.049	1.1248	1.0878
10	0.4361	0.3997	0.4483	0.0312	0.2181	0.4517	0.0519	1.1299	1.0909
---	---	---	---	---	---	---	---	---	---
100	0.8663	0.5357	0.6011	0.0277	-1.1169	0.6248	0.0891	1.1663	1.617

Hyperparameters:

Rank 1: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.016341160918740744, 'reg__l2_leaf_reg': 20.392989855781266, 'reg__bagging_temperature': 3.7538379565212088, 'reg__random_strength': 4.352822035906965, 'reg__subsample': 0.6941324214526934, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8051423454798032, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8778.938198885242}

Rank 2: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01599514857635919, 'reg__l2_leaf_reg': 21.435460094503323, 'reg__bagging_temperature': 3.7040610268033953, 'reg__random_strength': 4.332869114235704, 'reg__subsample': 0.6448251091619241, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.7988262184014666, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8626.282093803}

Rank 3: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.015600749519685035, 'reg__l2_leaf_reg': 23.359891801813973, 'reg__bagging_temperature': 3.678883634433316, 'reg__random_strength': 4.242444142092335, 'reg__subsample': 0.644745325347671, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7997689445414516, 'reg__border_count': 48, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9290.209216243142}

Rank 4: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.01496220040234218, 'reg__l2_leaf_reg': 20.859603087311978, 'reg__bagging_temperature': 3.581165925717731, 'reg__random_strength': 4.219273948576175, 'reg__subsample': 0.6435386250745411, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.787795900929367, 'reg__border_count': 49, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9844.873485739297}

Rank 5: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.014158368900892407, 'reg__l2_leaf_reg': 24.560107912036308, 'reg__bagging_temperature': 3.6389290380781487, 'reg__random_strength': 4.093576400772062, 'reg__subsample': 0.7511213305943281, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 10, 'reg__rsm': 0.8073828778752242, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8968.738700151289}

Rank 6: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.018929969279920522, 'reg__l2_leaf_reg': 17.504260898613307, 'reg__bagging_temperature': 4.029002565301298, 'reg__random_strength': 3.785810430190718, 'reg__subsample': 0.6476549182343799, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7745953285339712, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9452.773987066672}

Rank 7: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017015842293475972, 'reg__l2_leaf_reg': 20.64753444684127, 'reg__bagging_temperature': 3.6895074951256146, 'reg__random_strength': 4.389338912173276, 'reg__subsample': 0.6910350345317529,

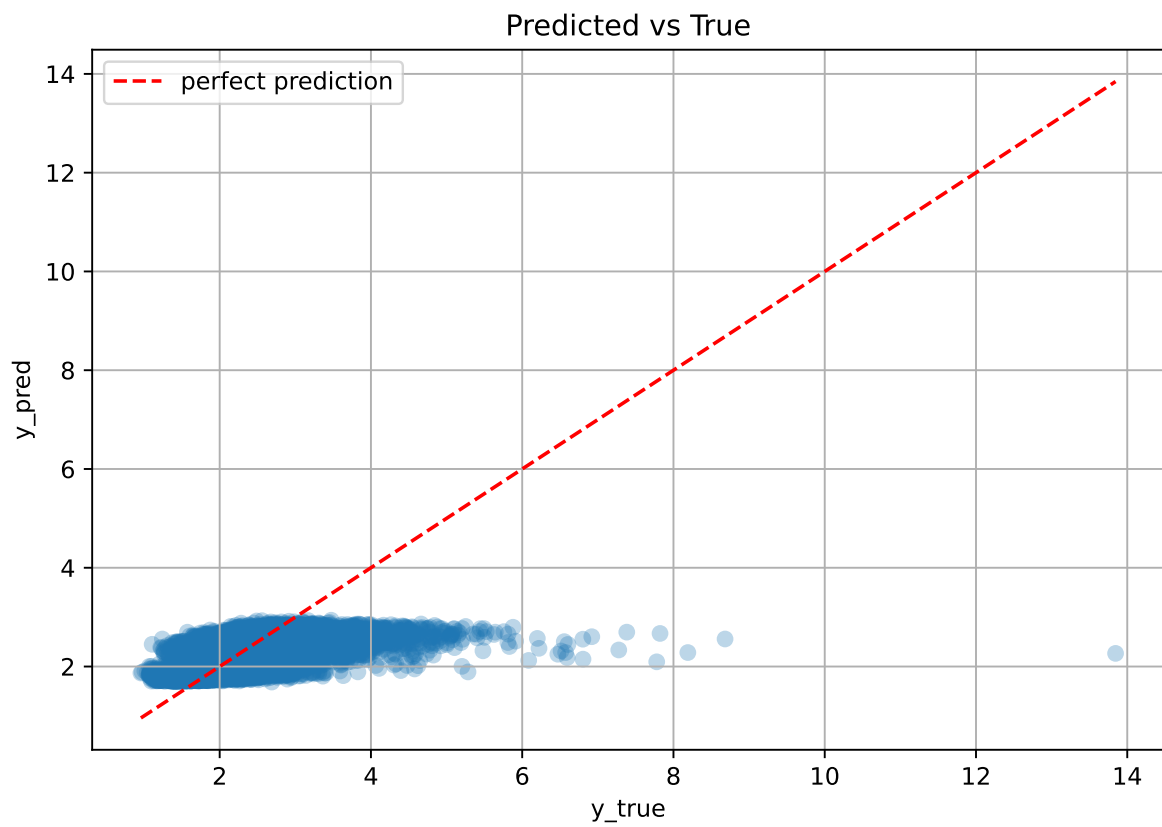
'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.8063008900472043, 'reg__border_count': 39, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8810.109987171134}

Rank 8: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.0166881496217311, 'reg__l2_leaf_reg': 13.009748685138572, 'reg__bagging_temperature': 4.462203323936929, 'reg__random_strength': 3.368898439451301, 'reg__subsample': 0.5948760398069541, 'reg__min_data_in_leaf': 26, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7506569326902306, 'reg__border_count': 37, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8201.703783910258}

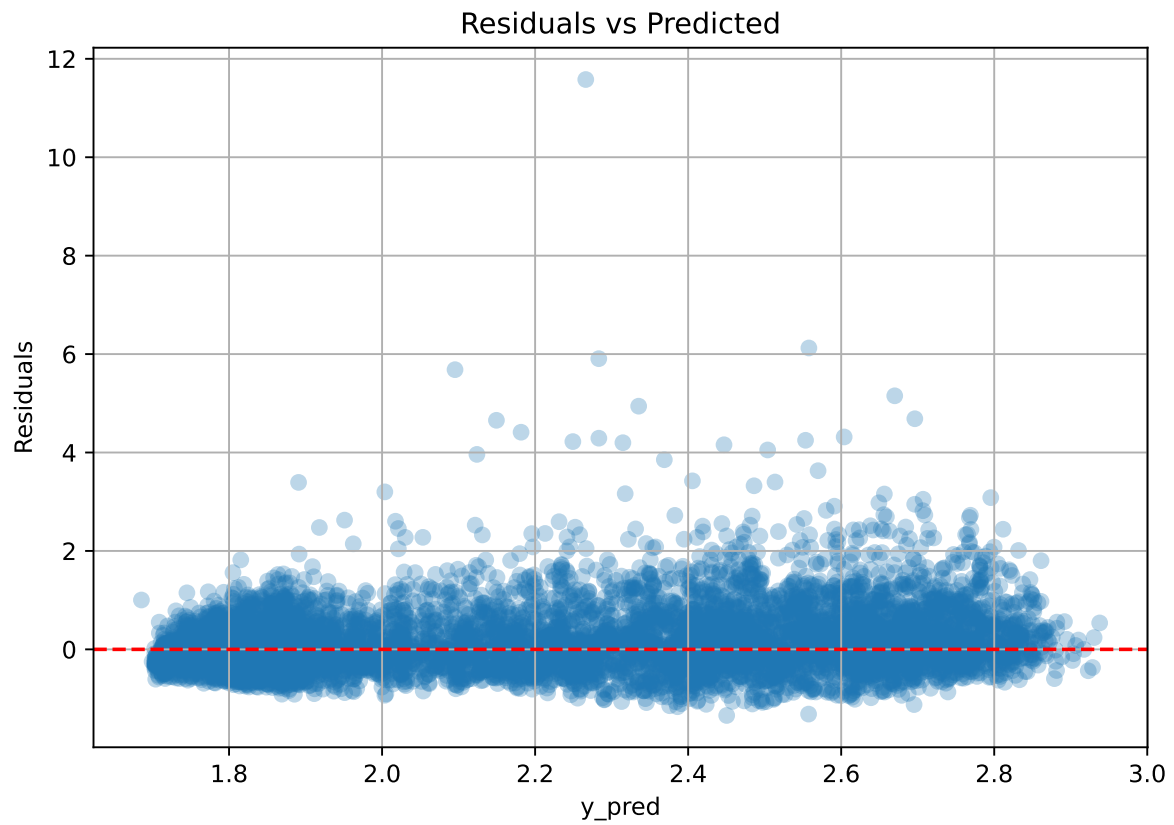
Rank 9: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.018403863483704375, 'reg__l2_leaf_reg': 19.51924753601154, 'reg__bagging_temperature': 4.169887119711394, 'reg__random_strength': 4.5109656655170225, 'reg__subsample': 0.6448527241486087, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7752557246495136, 'reg__border_count': 35, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8464.816572928483}

Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.019203791690250077, 'reg__l2_leaf_reg': 18.29034393859382, 'reg__bagging_temperature': 4.141676251177872, 'reg__random_strength': 3.8747305396474605, 'reg__subsample': 0.6194141825352872, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7757179408632854, 'reg__border_count': 34, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8383.681773290507}

2.4.6 Scatter



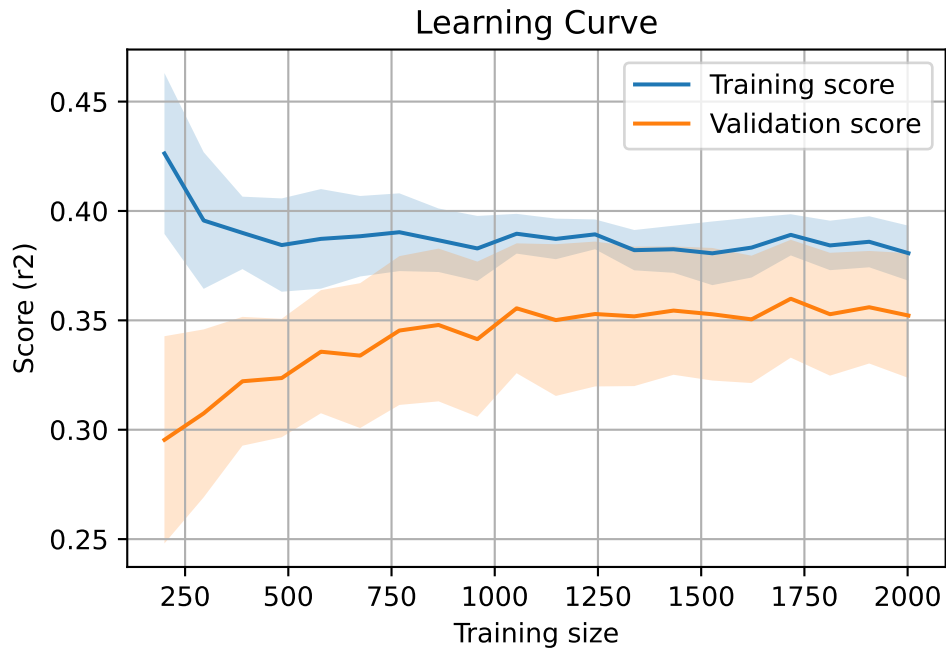
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 270

`Len(group)` : 4086, `Len(training)` : 2861,

`Len(training_real)` : 2003, `Len(test_of_training)` : 858, `Len(test)` : 1225



2.5 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	test_cv	std_test_cv	pen_score	test	diff	r_test	r_train
LGBM	155	2858	1224	4082	0.545	0.519	0.56	0.06	0.5192	0.584	0.065	1.126	1.05
LGBM	270	2861	1225	4086	0.316	0.293	0.332	0.026	0.1807	0.349	0.056	1.19	1.077
LGBM	MEAN_WEIGHTED	5719	2449	8168	0.43	0.406	0.446	0.043	0.35	0.467	0.061	1.158	1.063
CatBoost	155	2858	1224	4082	0.608	0.581	0.623	0.059	0.5806	0.63	0.049	1.084	1.048
CatBoost	270	2861	1225	4086	0.385	0.359	0.407	0.029	0.2296	0.406	0.047	1.132	1.072
CatBoost	MEAN_WEIGHTED	5719	2449	8168	0.497	0.47	0.515	0.044	0.405	0.518	0.048	1.108	1.06

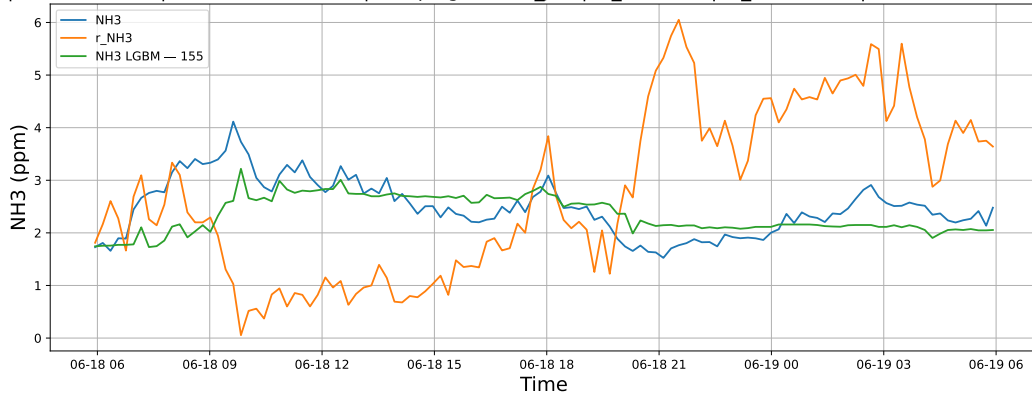
3 Plot

3.1 Standalone plots

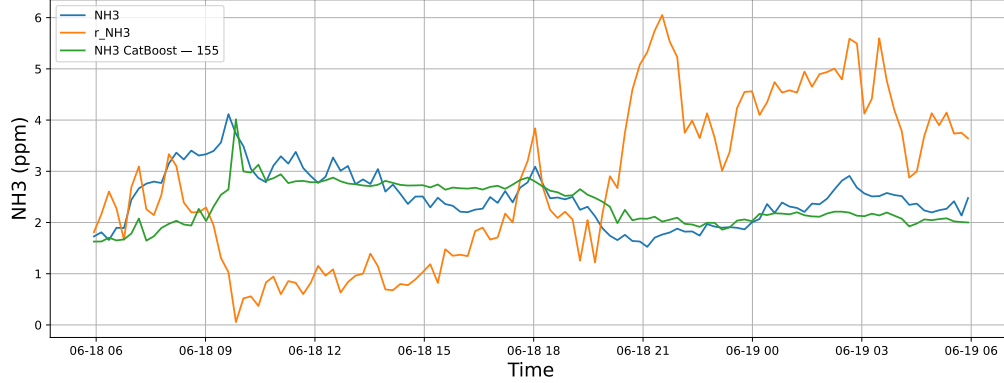
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

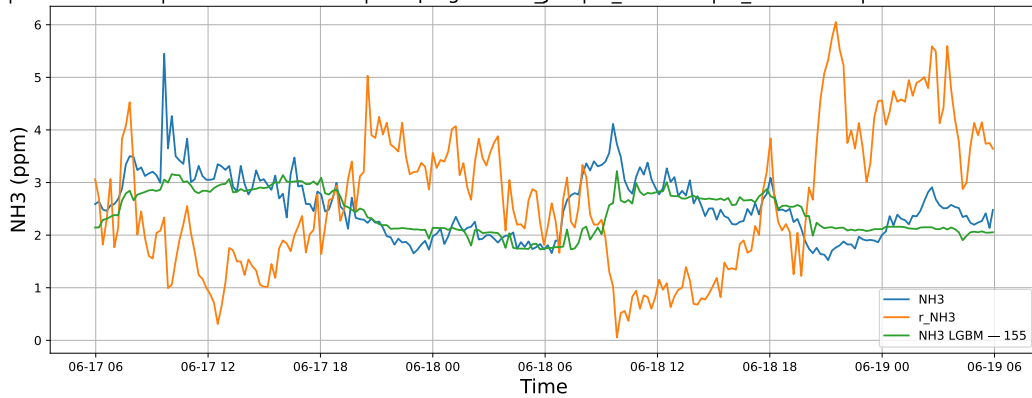


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

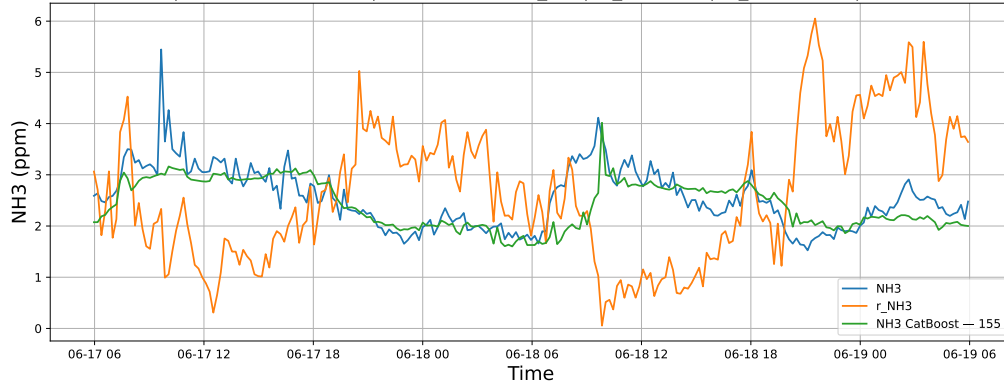


3.1.1.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

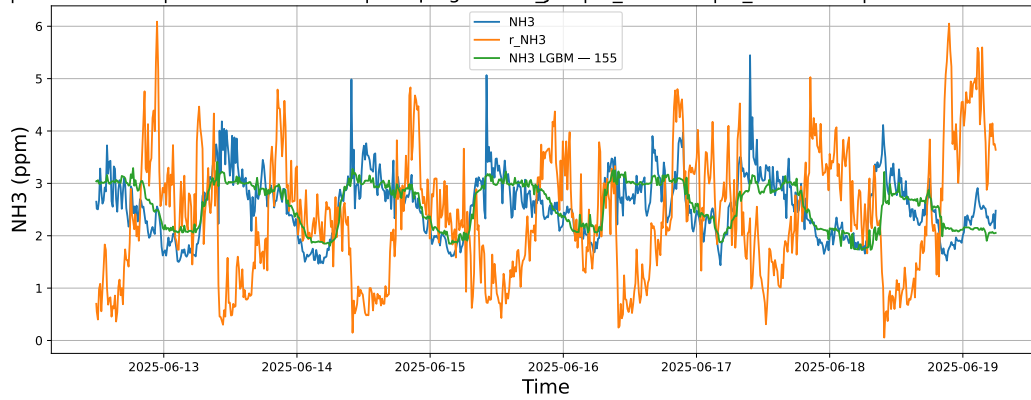


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

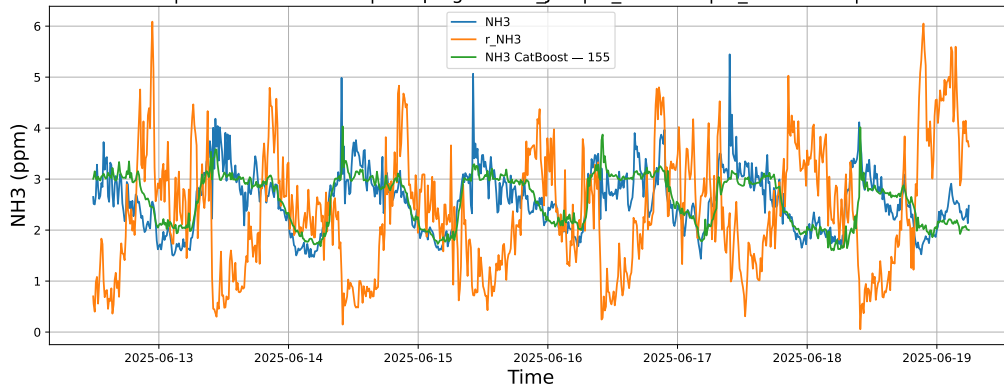


3.1.1.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



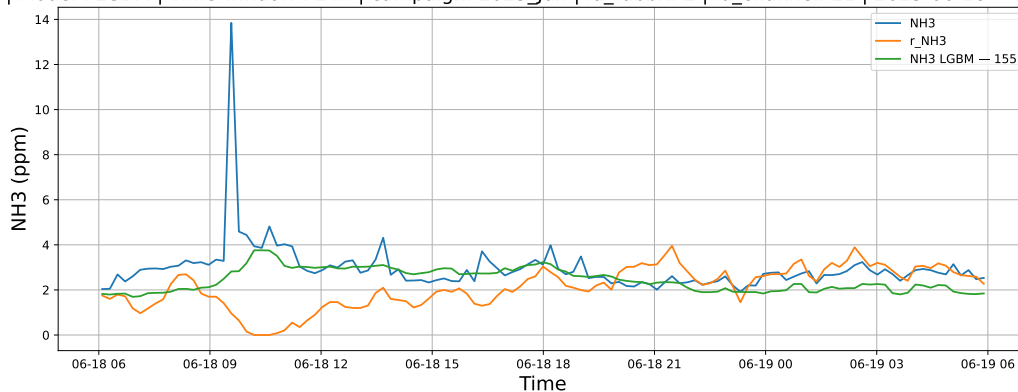
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



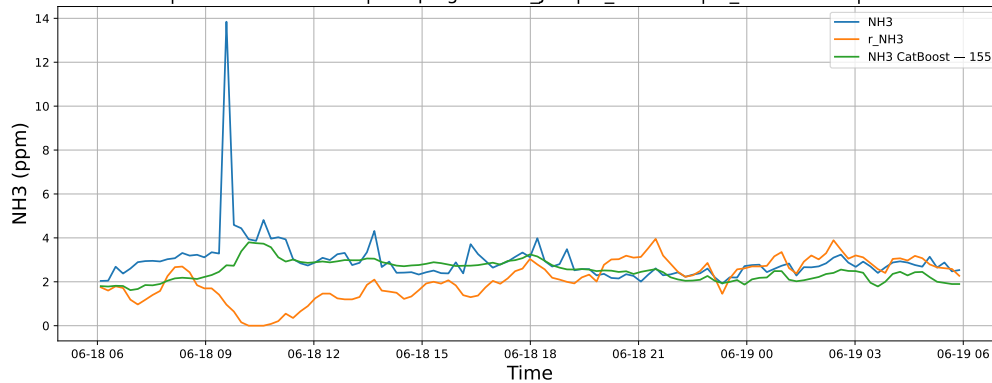
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

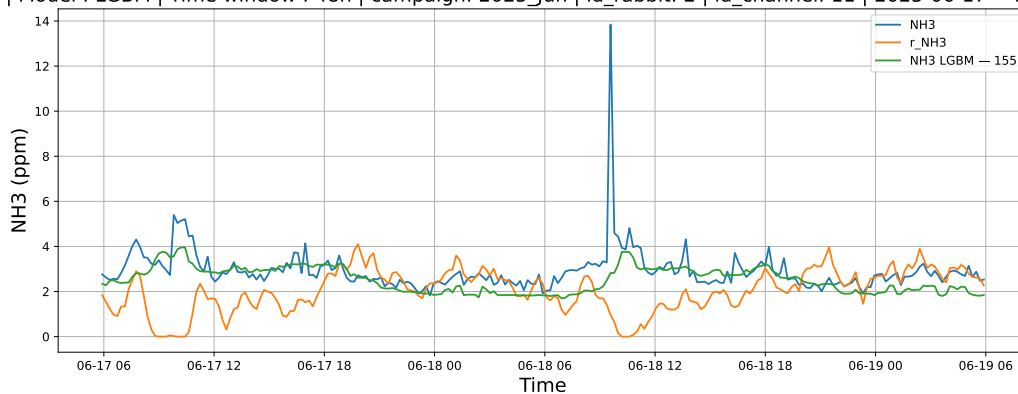


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

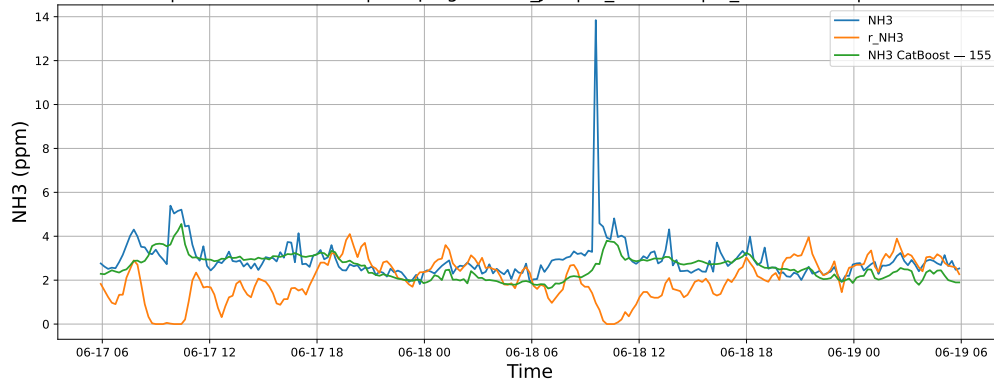


3.1.2.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

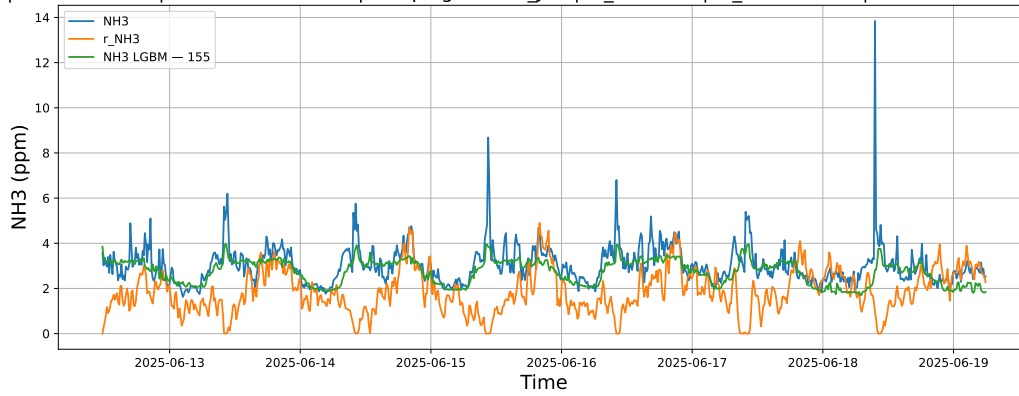


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

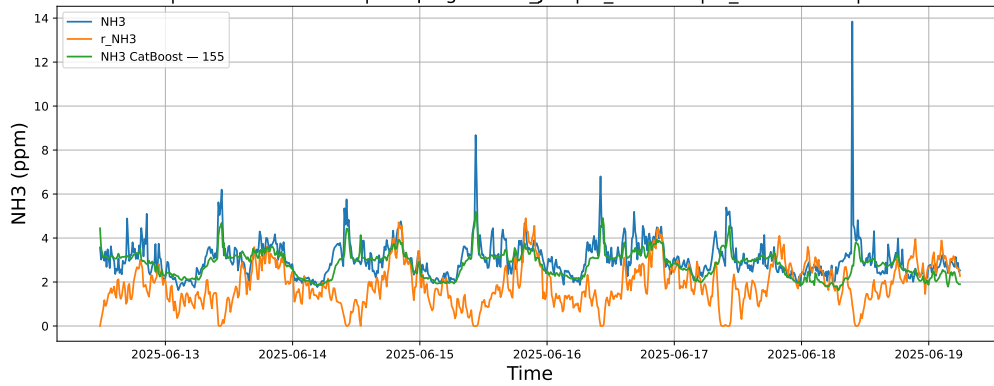


3.1.2.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



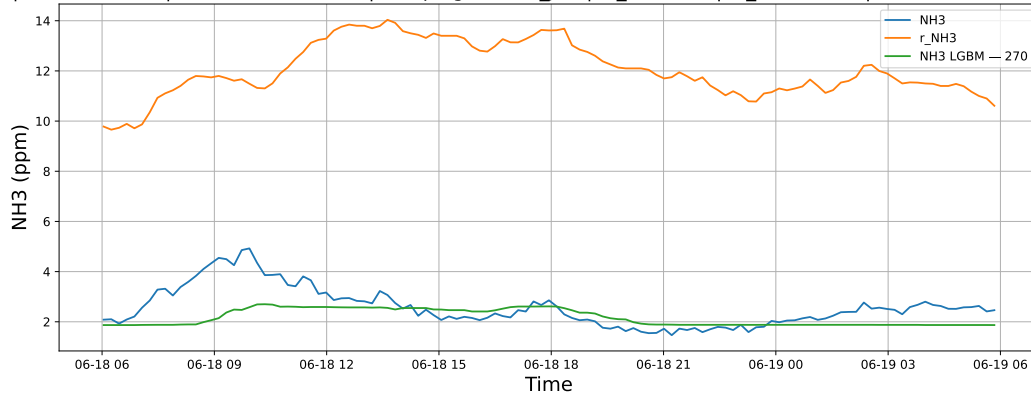
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



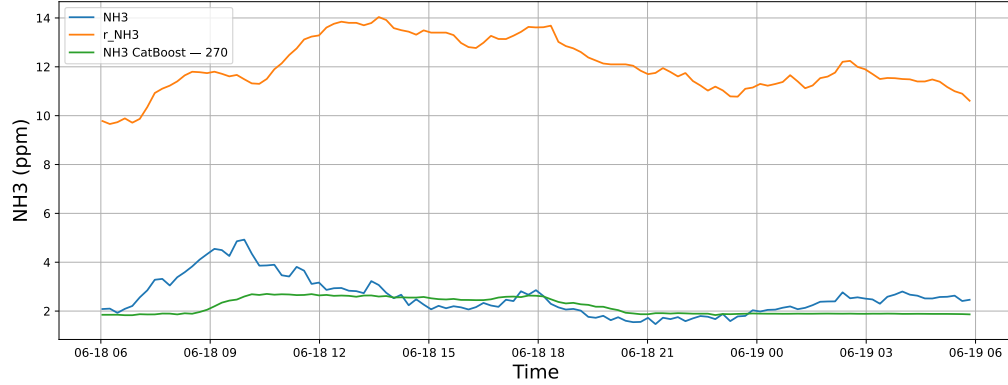
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

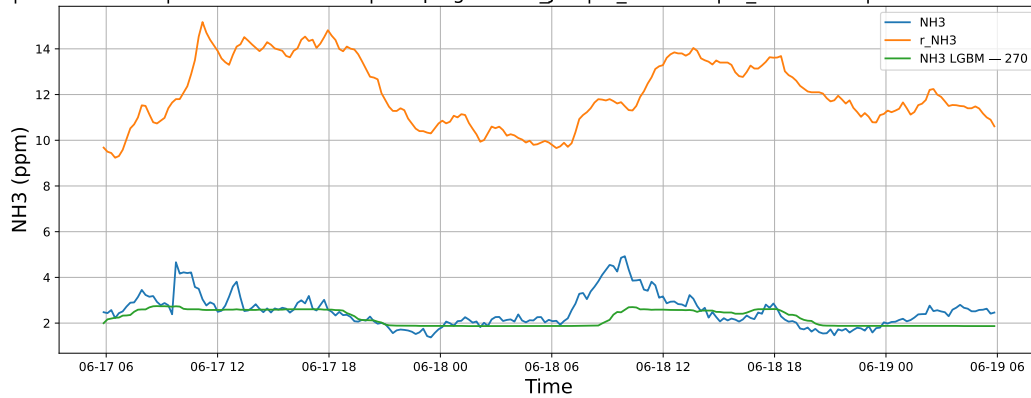


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

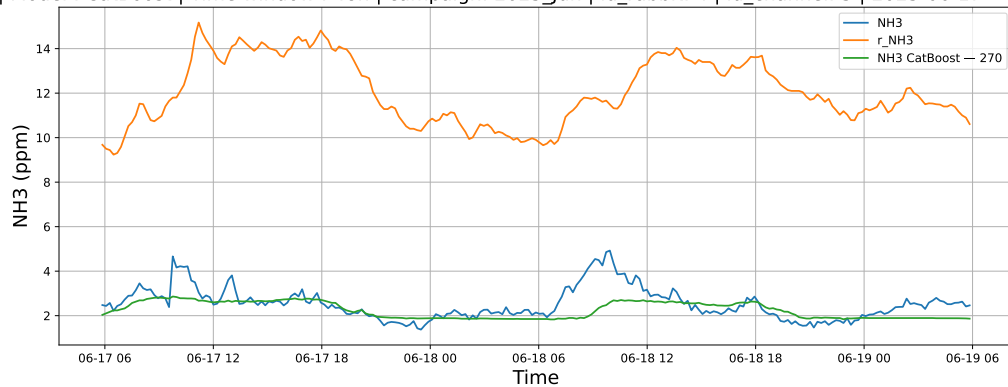


3.1.3.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

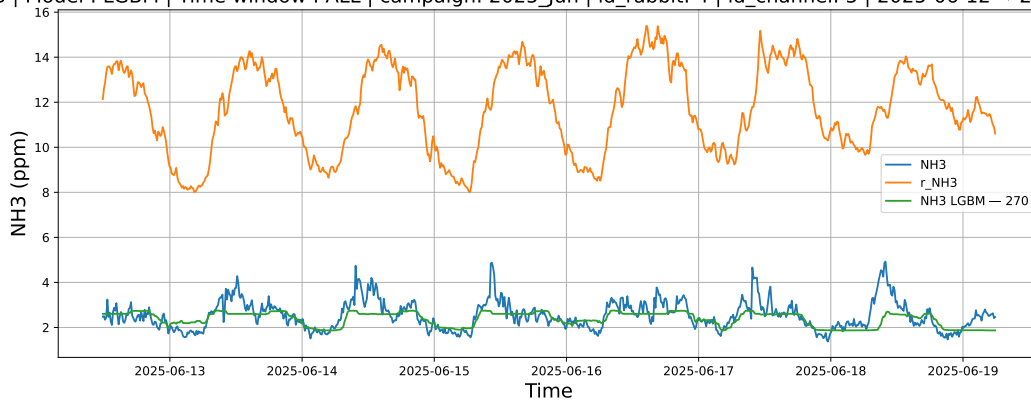


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

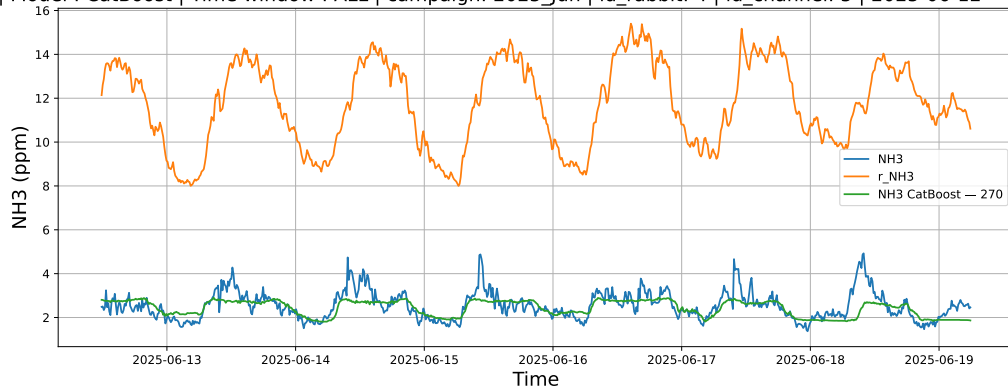


3.1.3.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



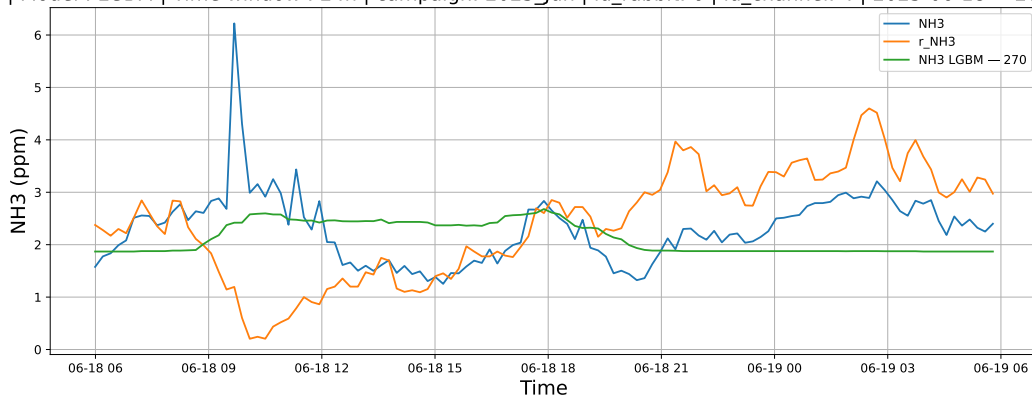
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

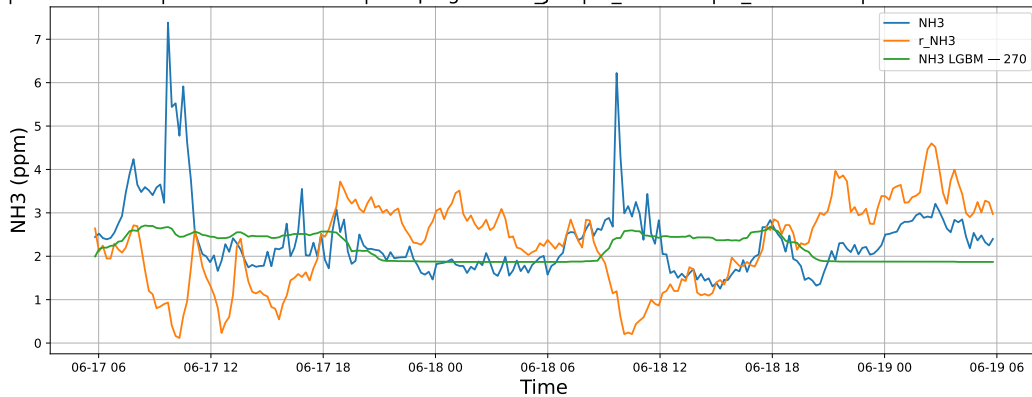


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

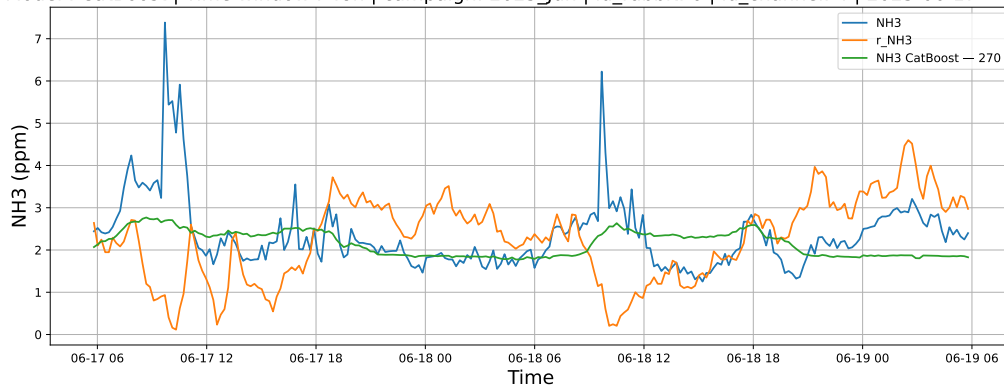


3.1.4.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

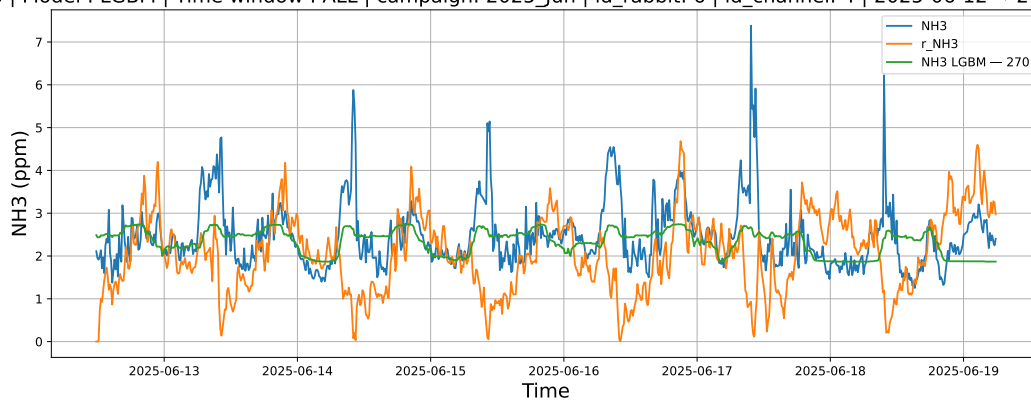


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

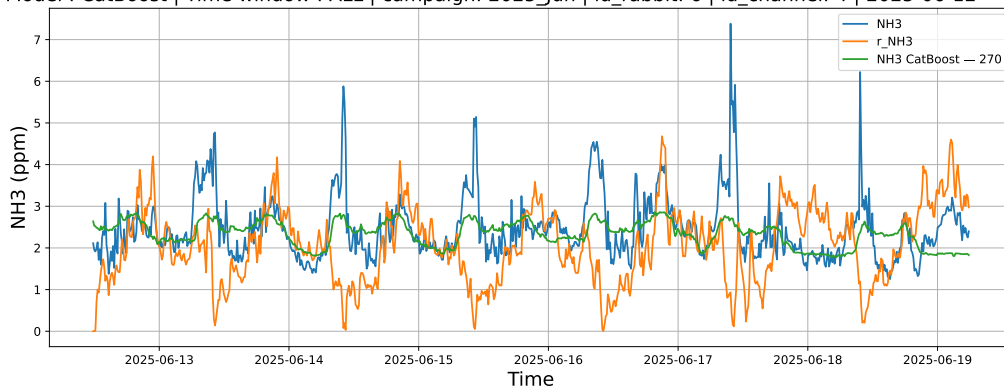


3.1.4.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



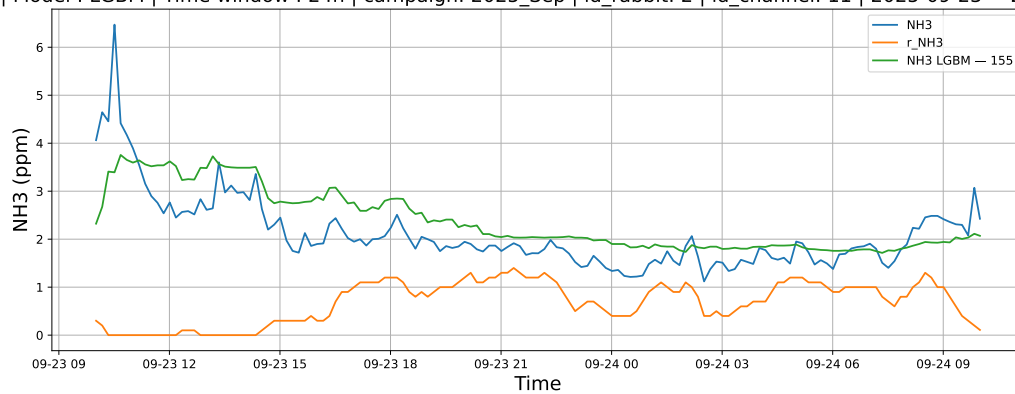
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



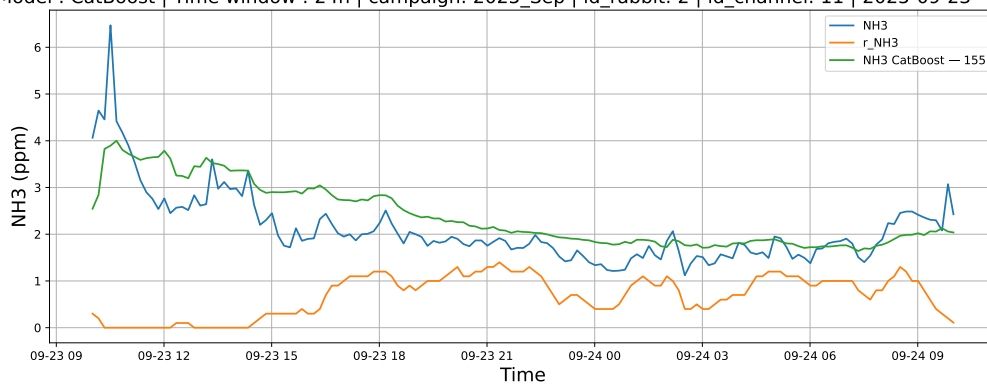
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

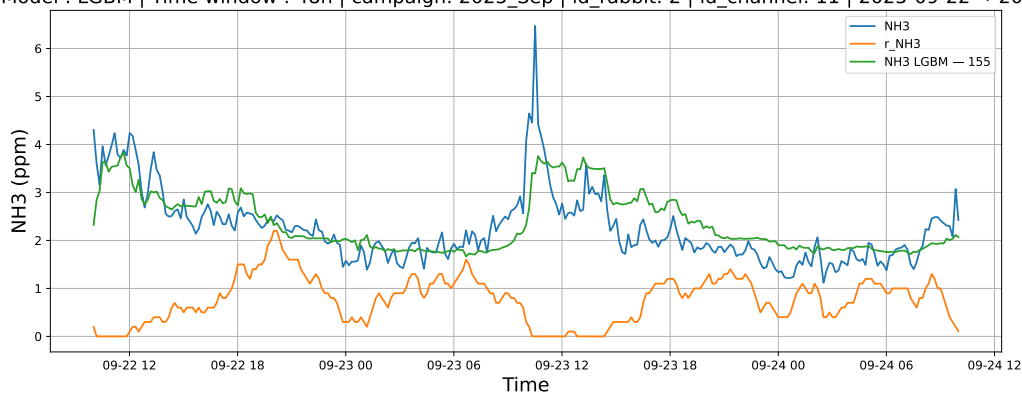


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

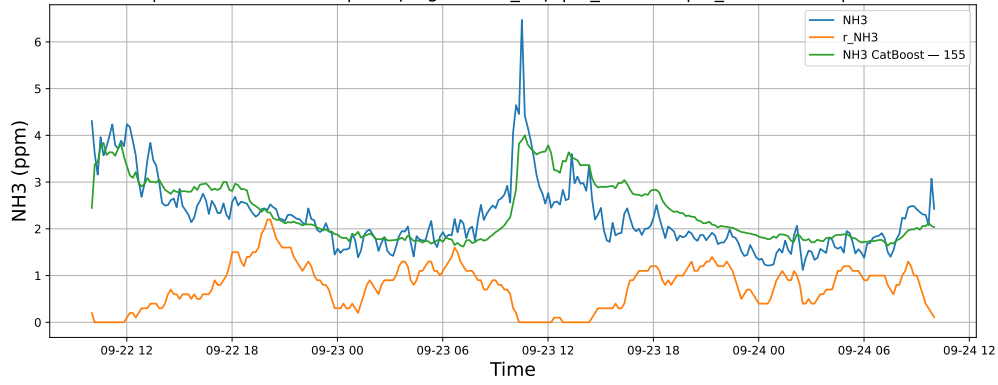


3.1.5.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

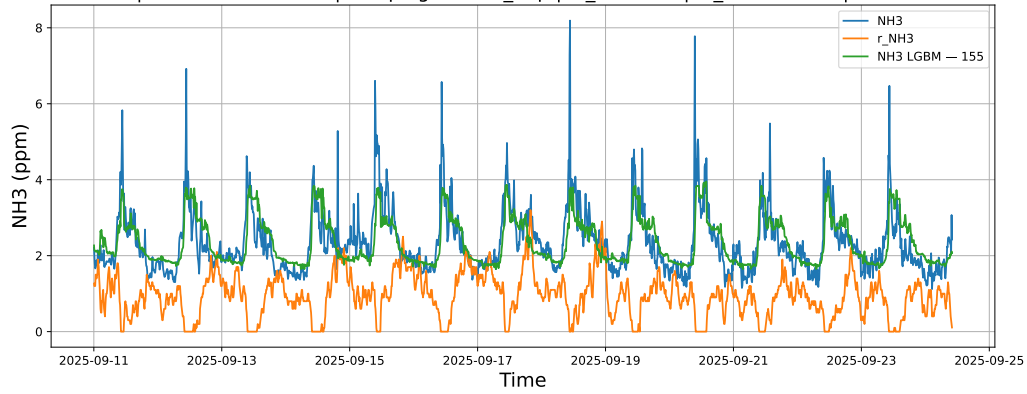


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

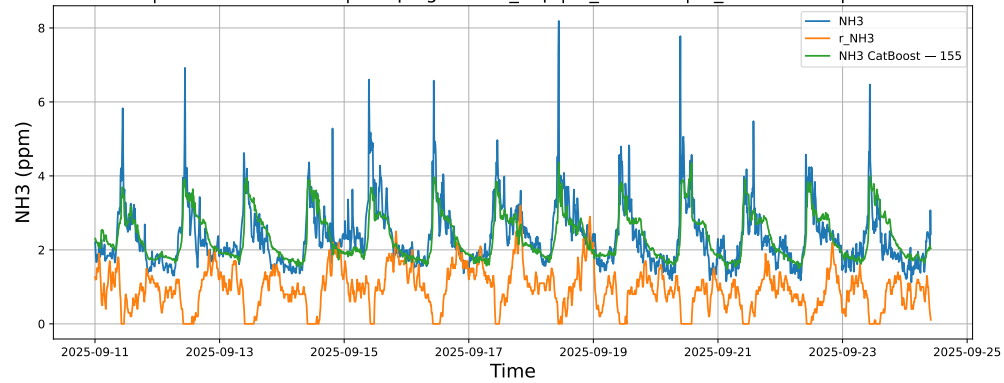


3.1.5.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



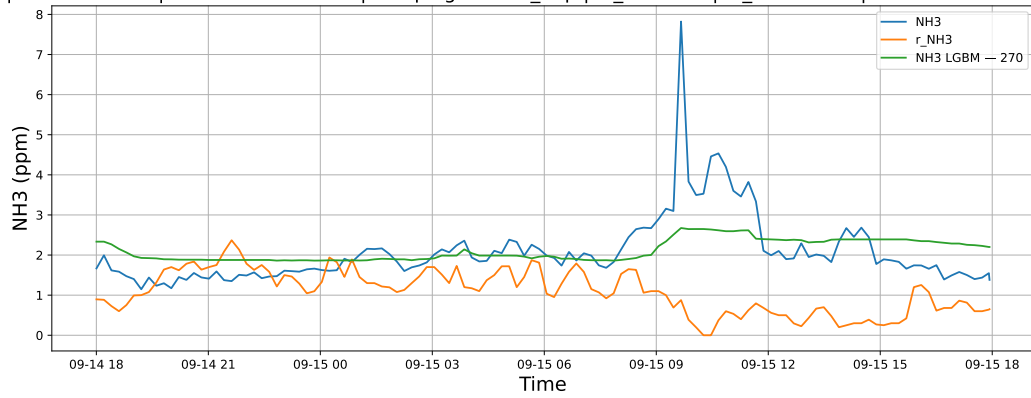
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



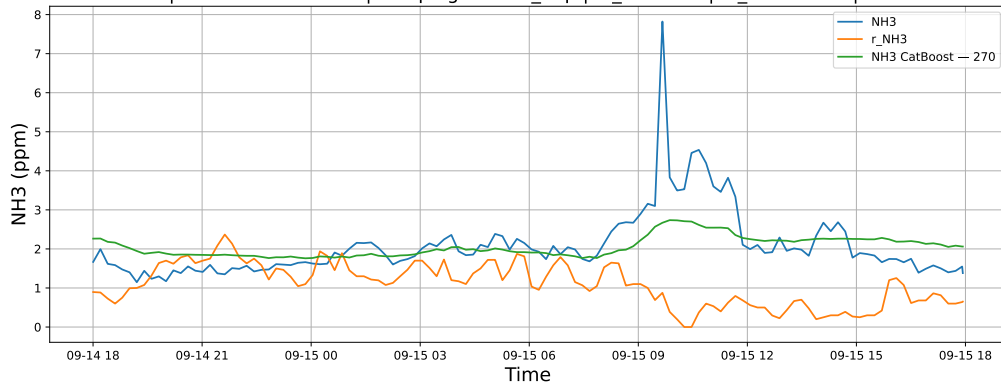
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

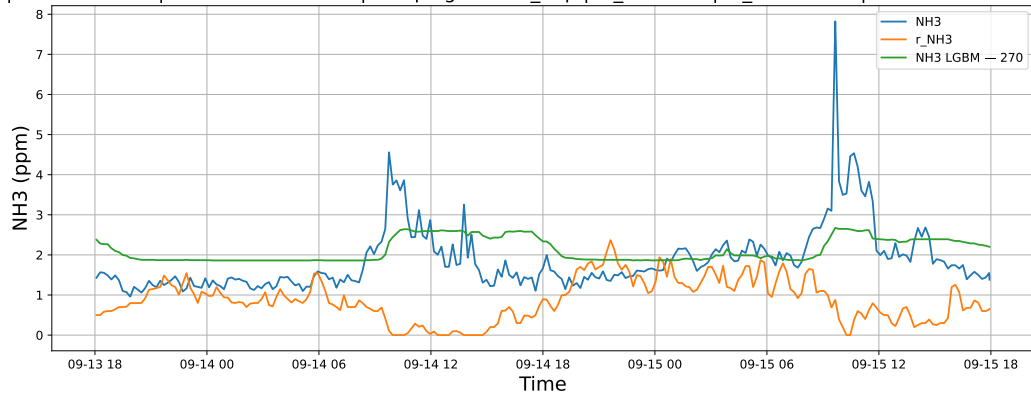


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

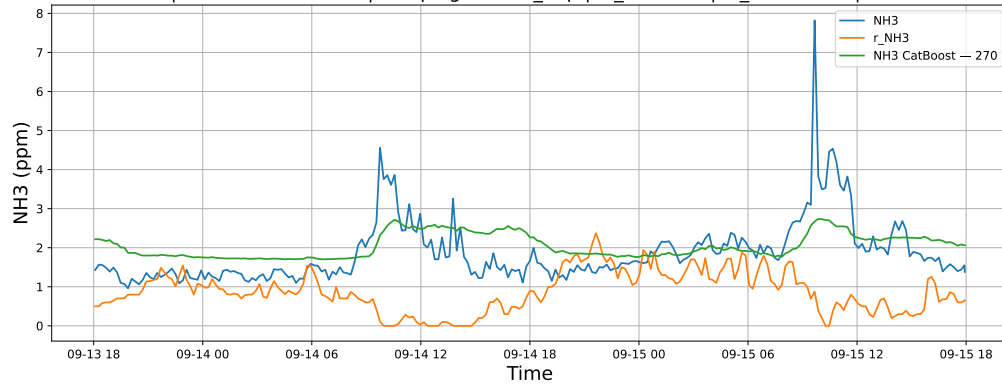


3.1.6.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

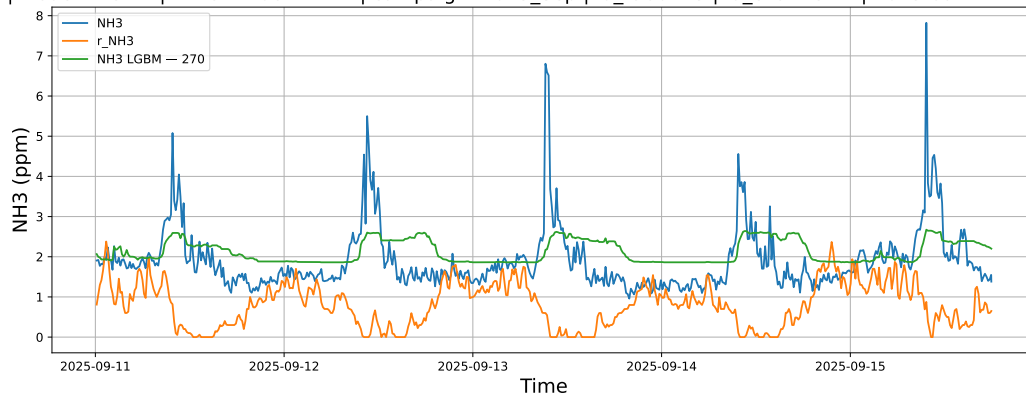


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

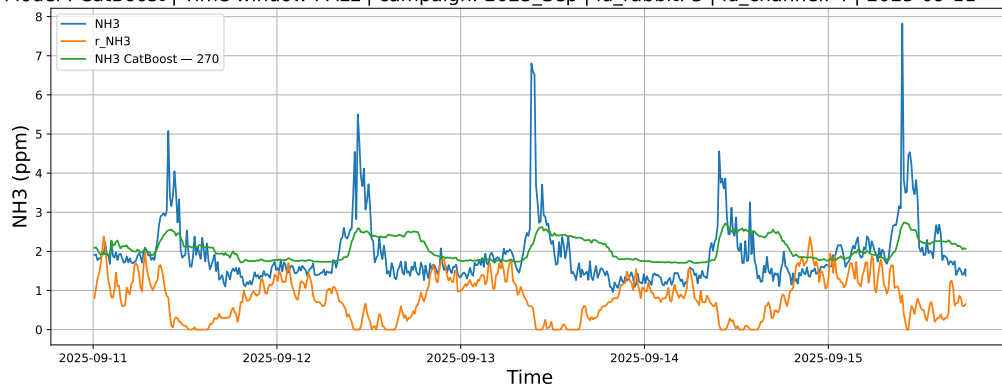


3.1.6.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



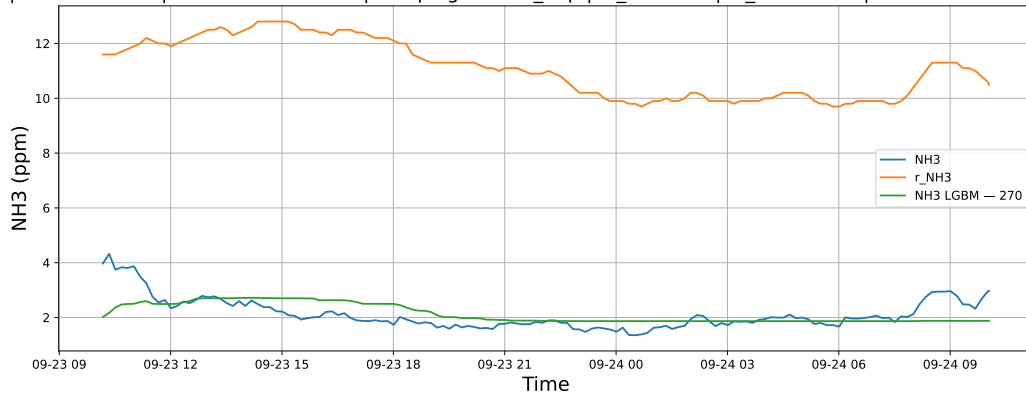
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



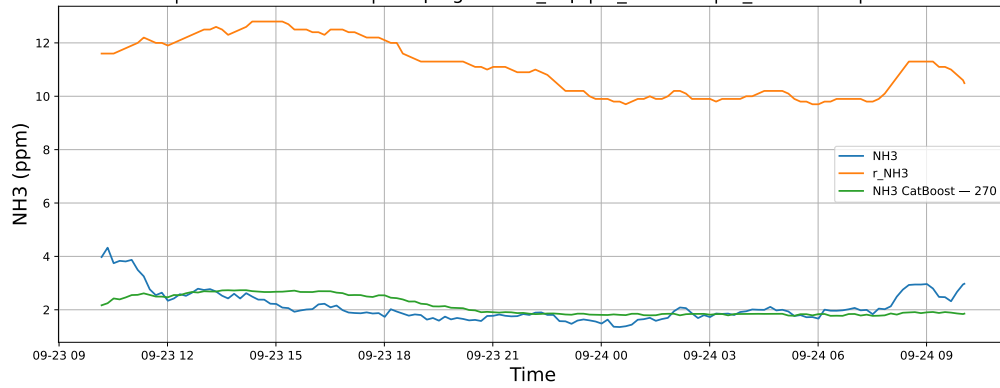
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

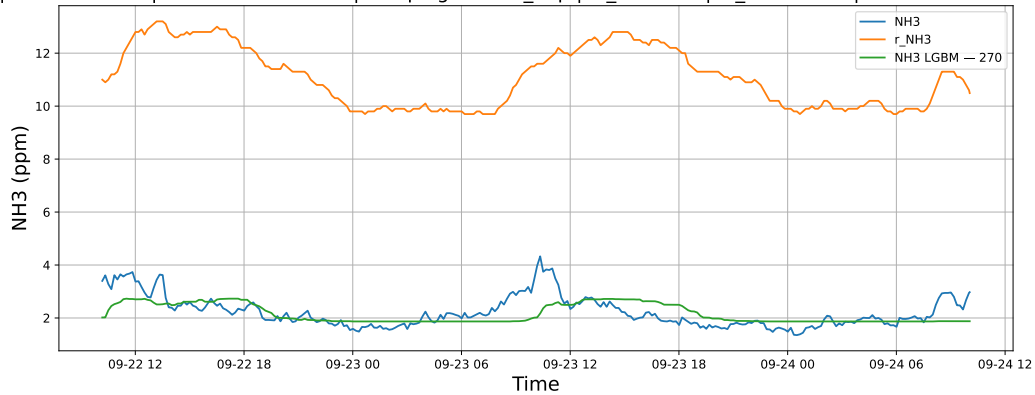


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

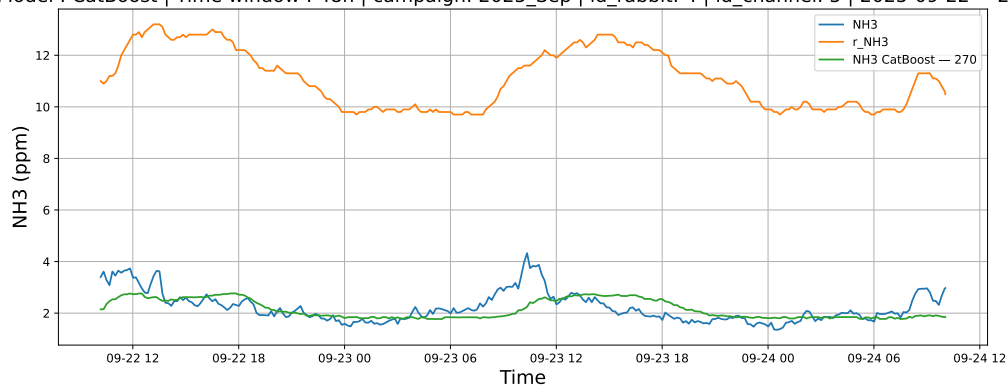


3.1.7.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

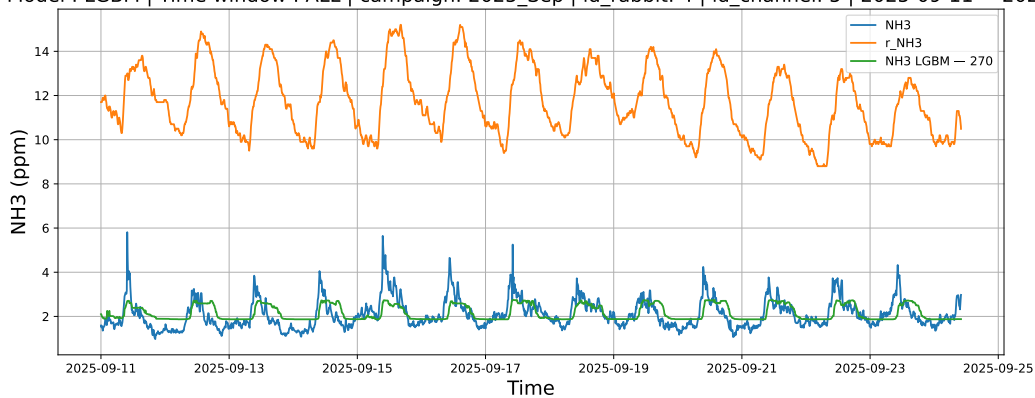


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

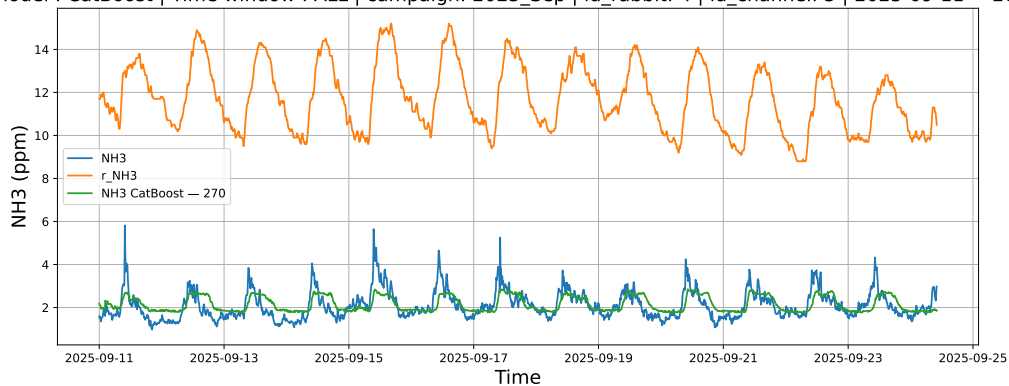


3.1.7.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



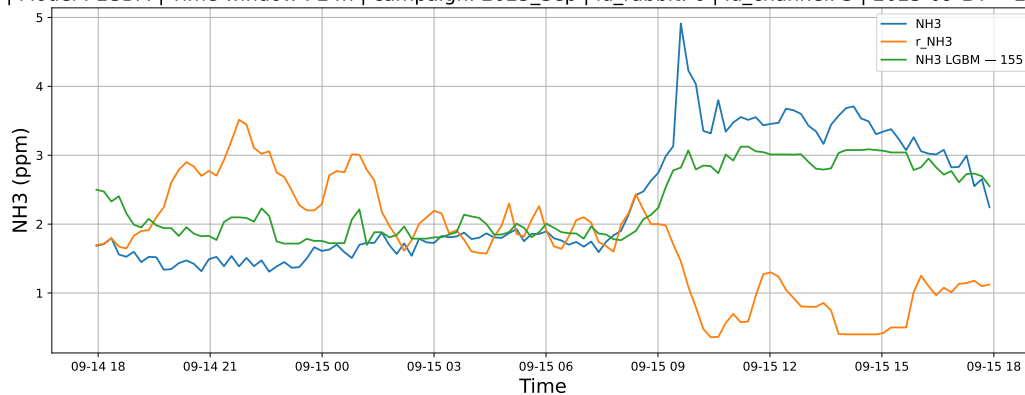
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



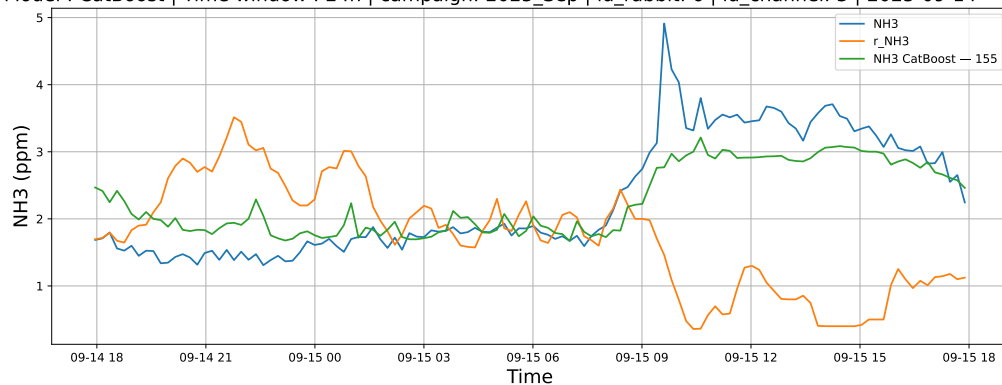
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

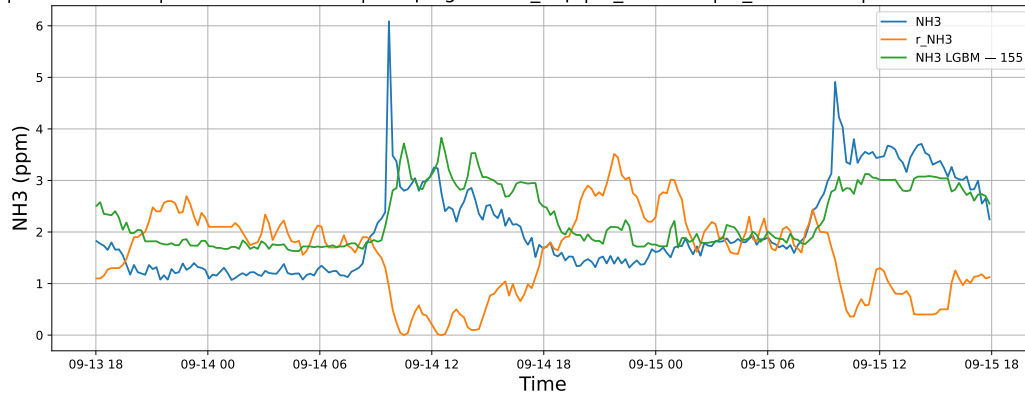


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

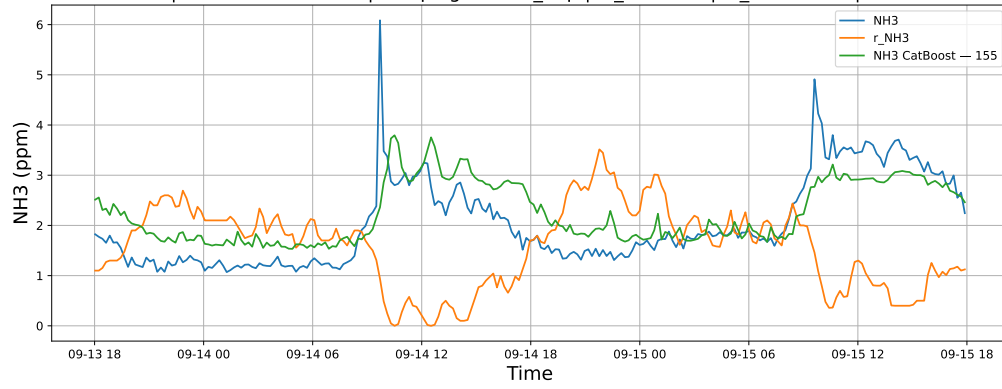


3.1.8.2 Time window : 48

NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

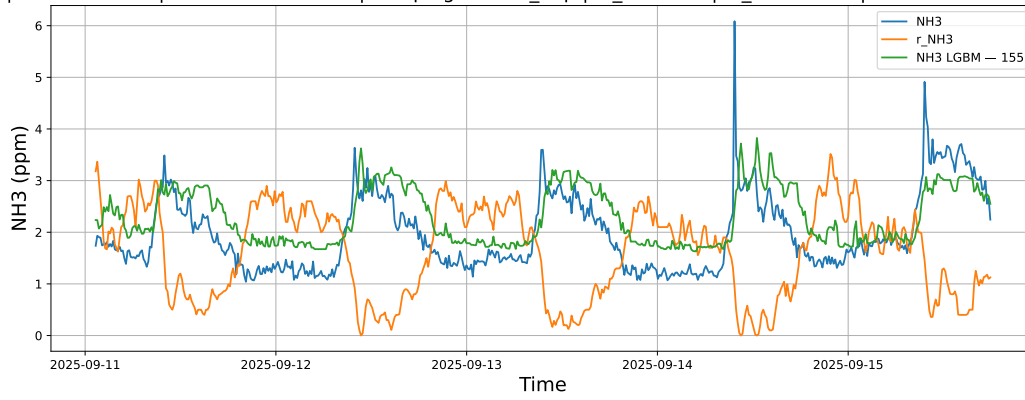


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

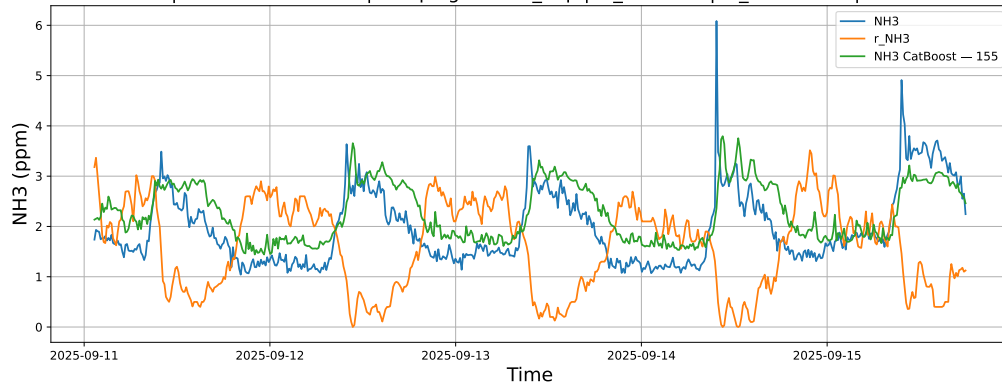


3.1.8.3 Time window : ALL

NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

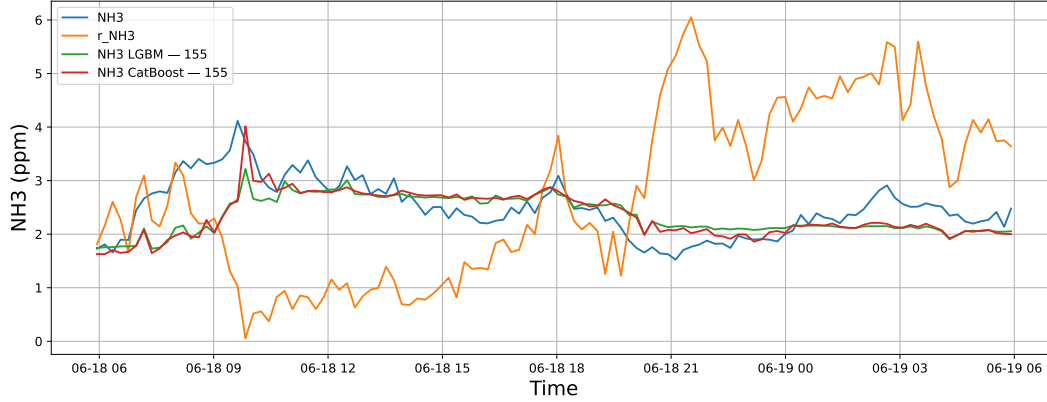


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

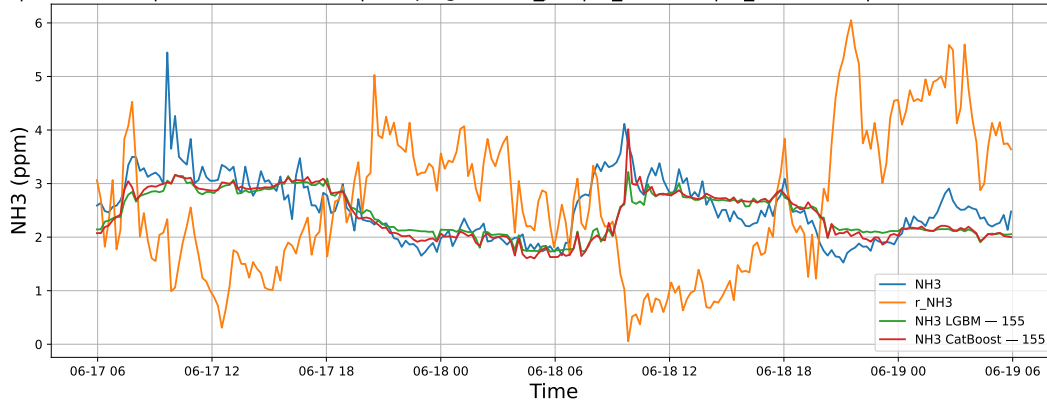
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



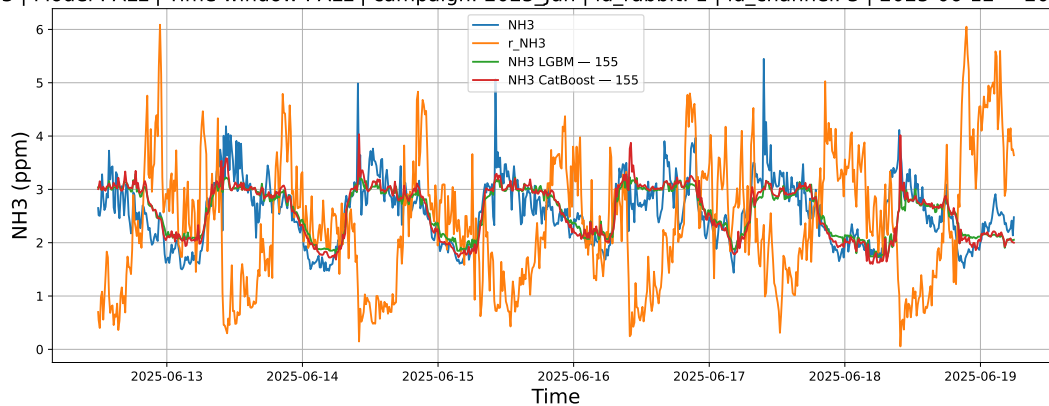
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

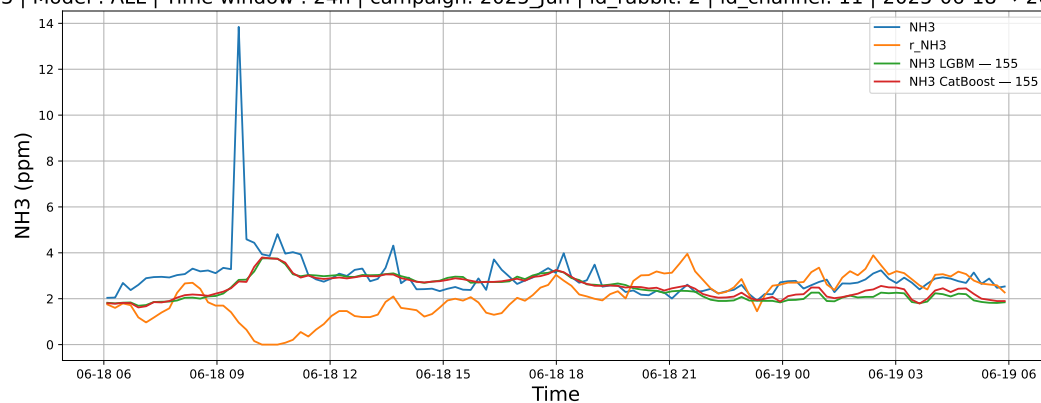
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

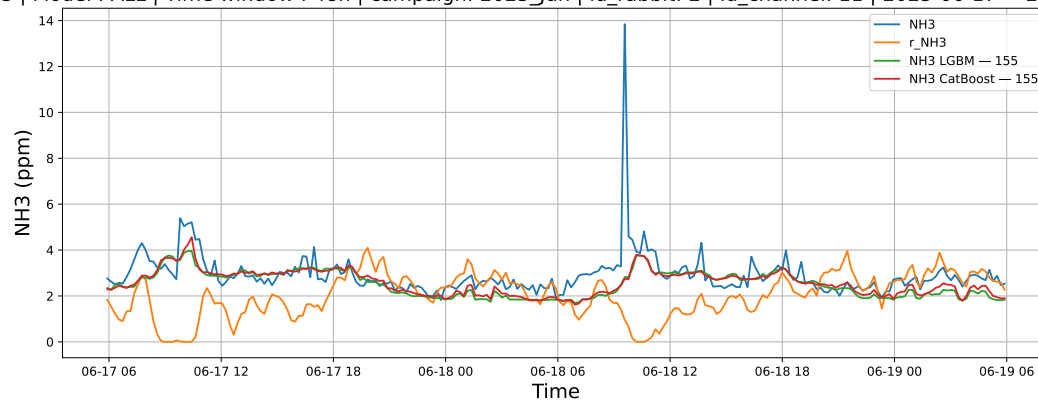
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



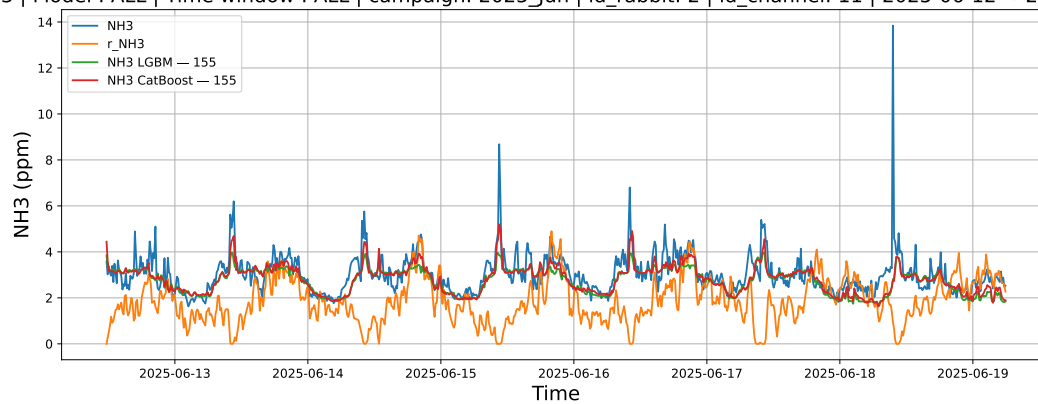
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

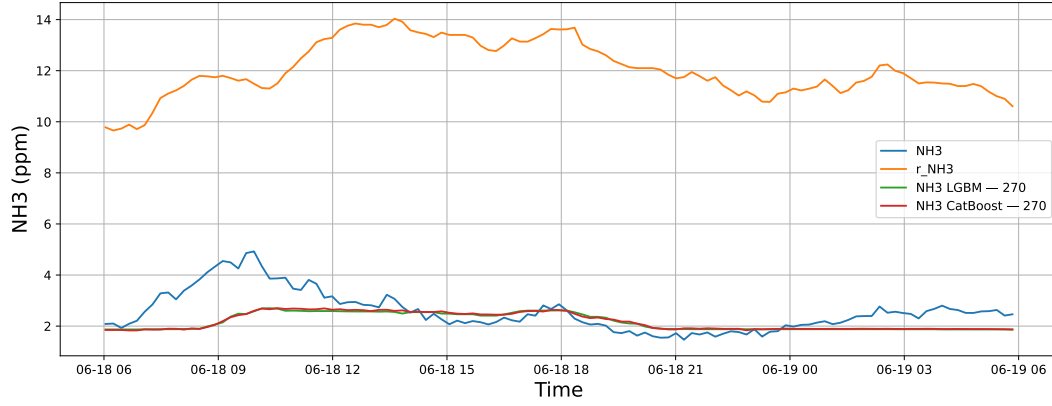
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

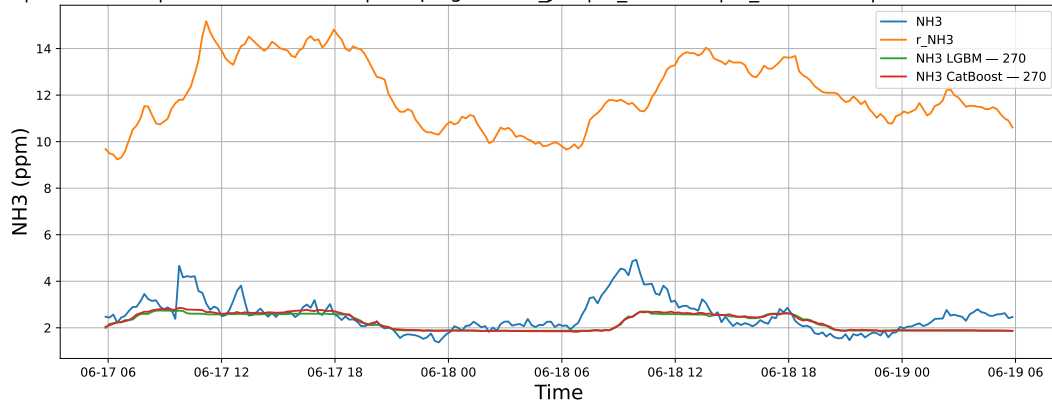
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



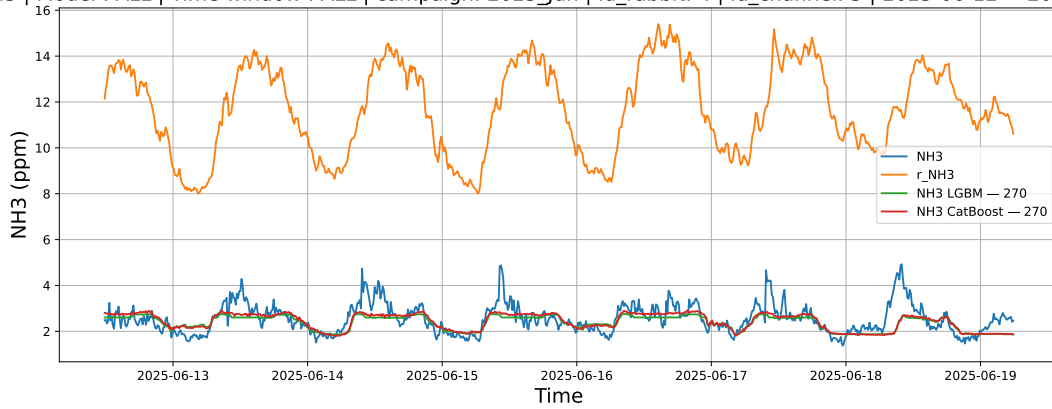
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

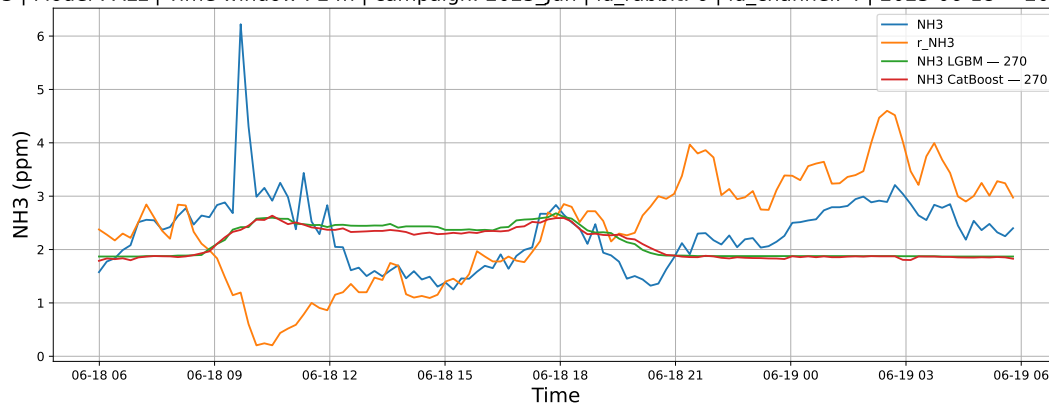
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

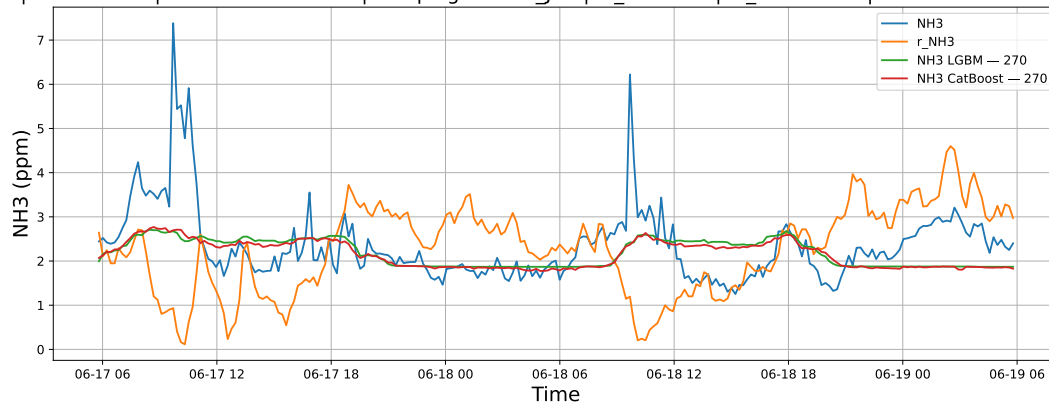
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



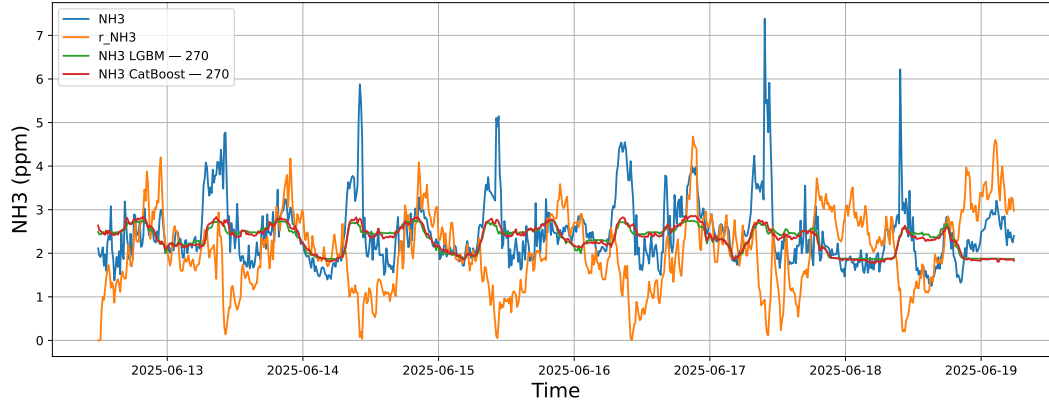
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

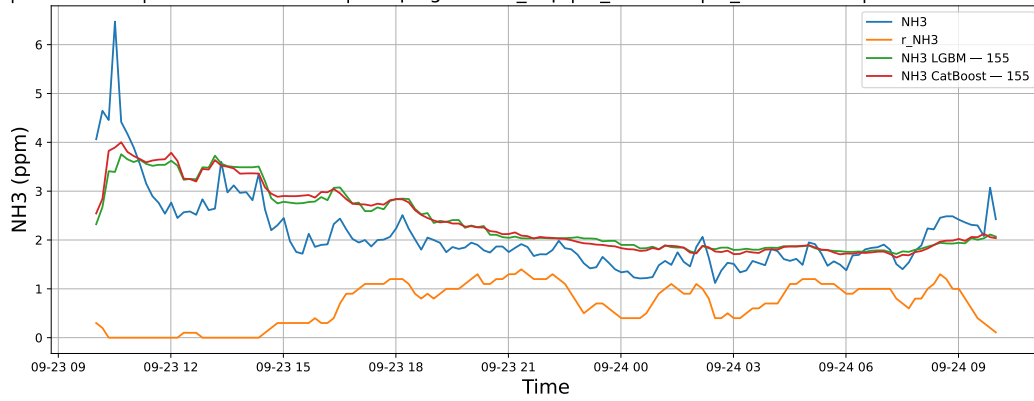
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

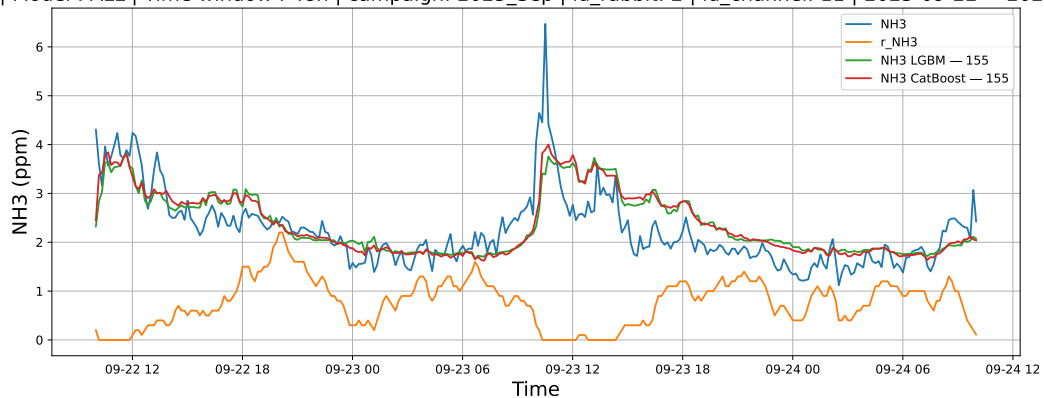
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



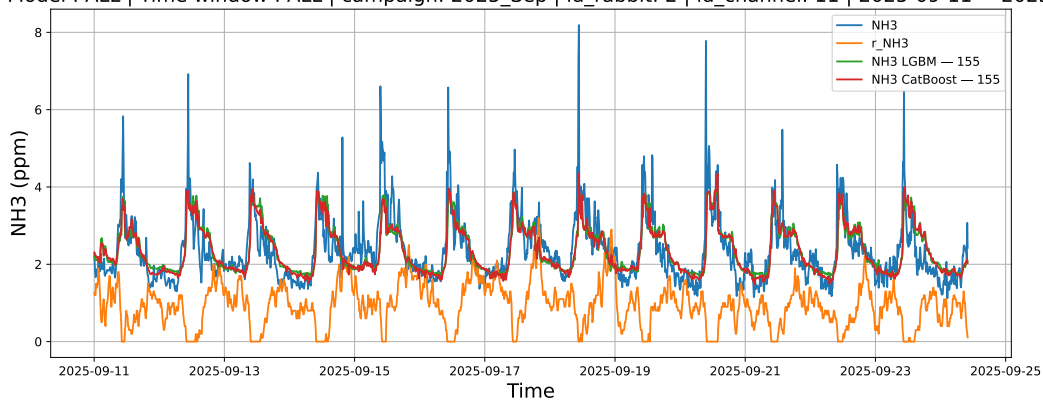
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

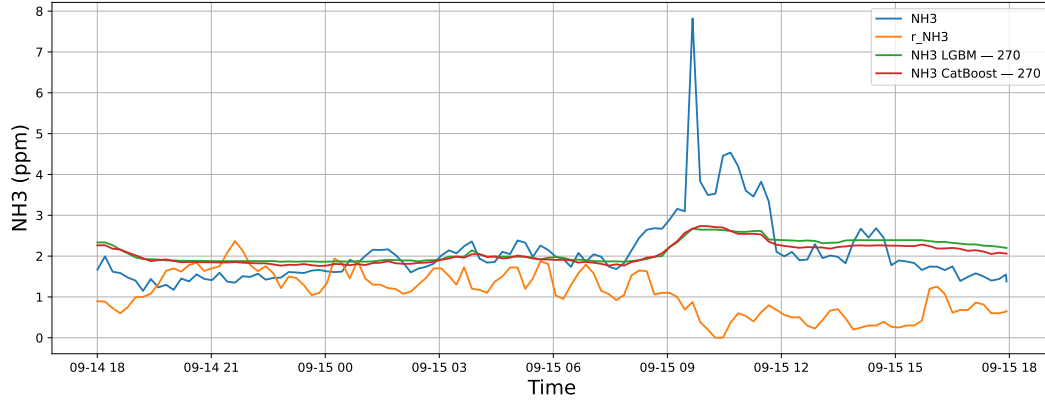
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

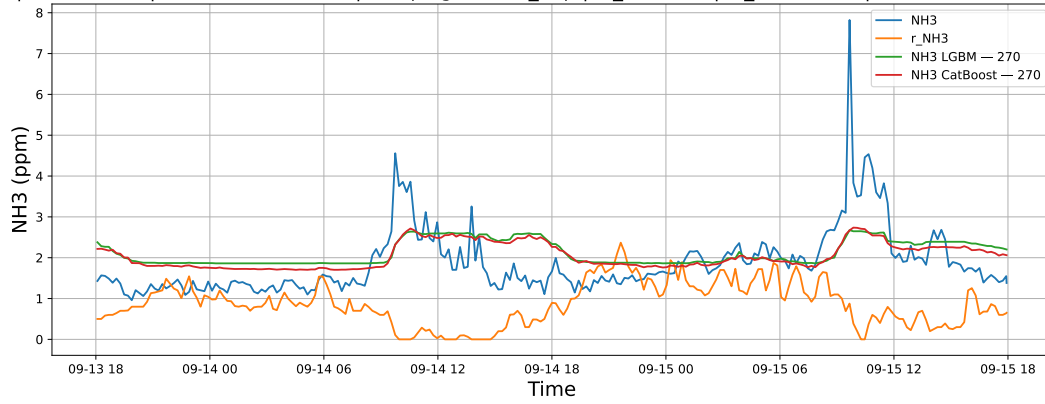
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



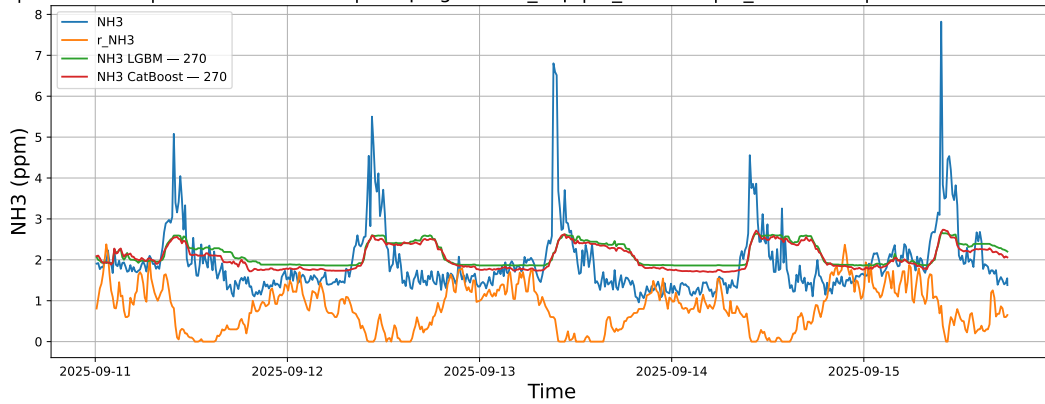
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

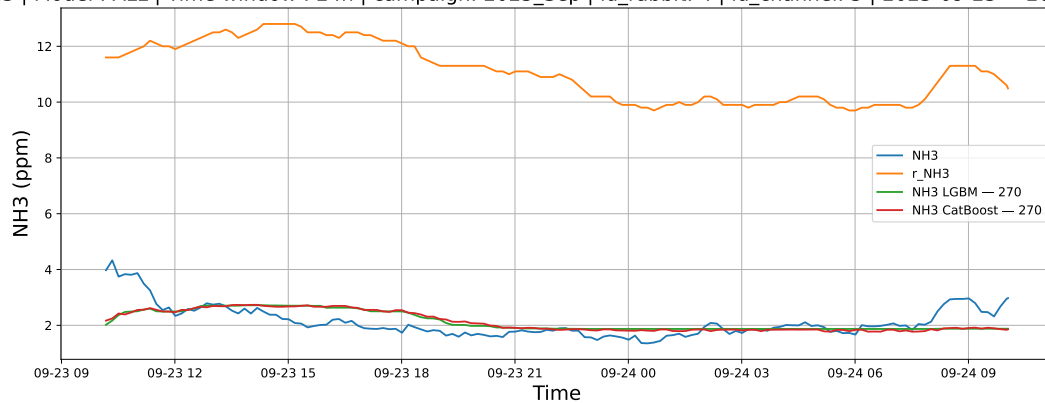
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

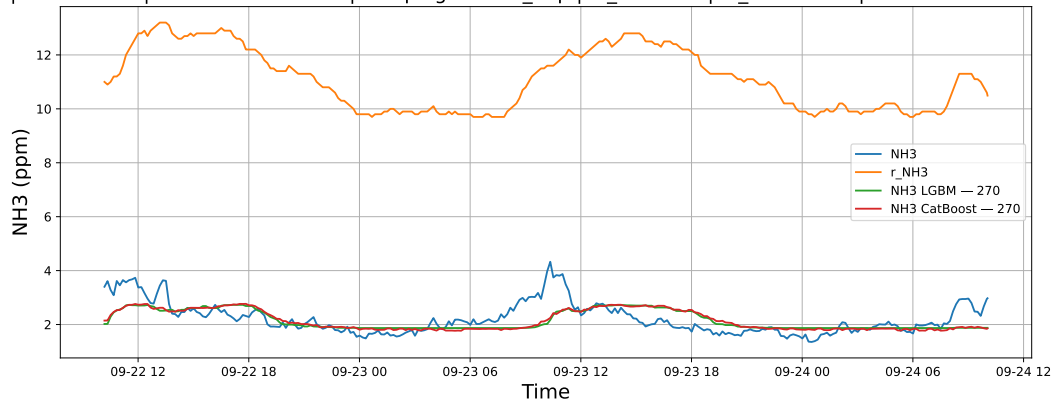
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



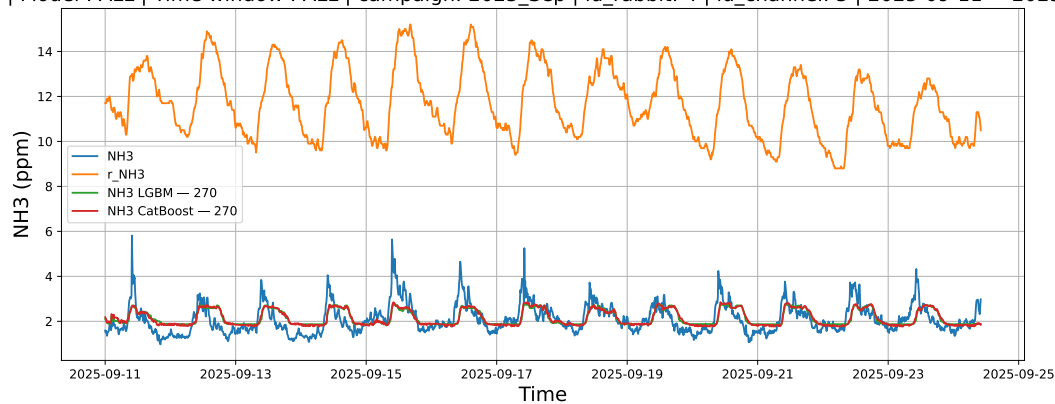
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

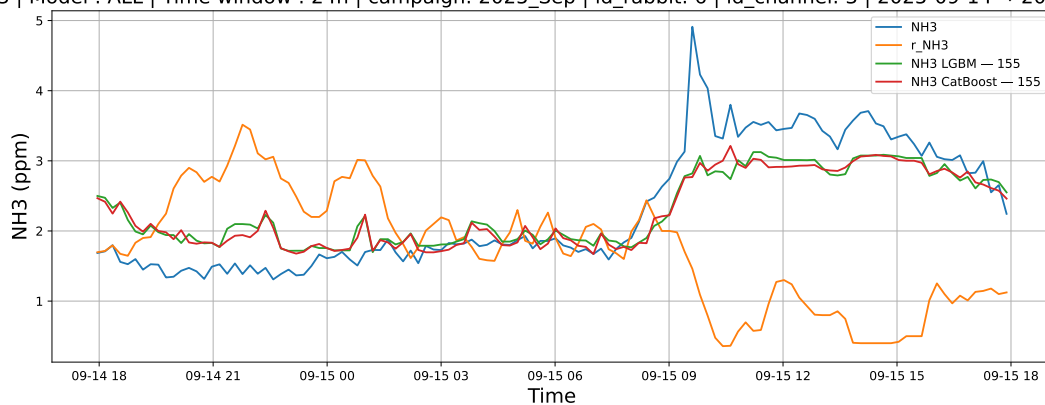
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

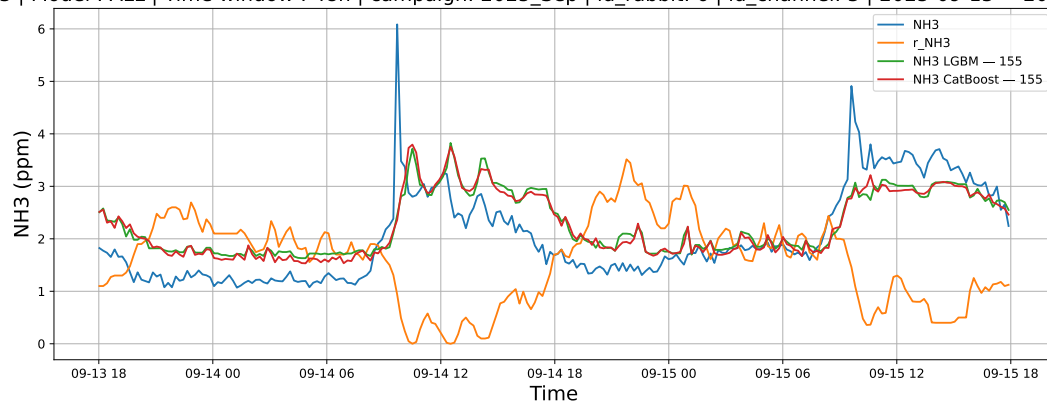
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

